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The Slate Islands meteorite impact site: a study of Shock Remanent Magnetization

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Summary. A detailed palaeomagnetic study of the Slate Islands meteorite impact site, northern Lake Superior, has successfully isolated a secondary component of remanence which on the basis of field relationships, models of impact crater formation and laboratory investigation appears to be a Shock Remanent Magnetization (SRM).

Evidence for this hitherto rare type of naturally formed remanence is that:

(a) the component is confined to rocks having experienced intense shock (50 to more than 100 kb) on the basis of shatter-cone development and the presence of planar features in quartz and felspar,
(b) the component appears to have been acquired virtually instantaneously in terms of normal geological processes as a combined palaeomagnetic–shatter cone analysis suggests that it was formed within a time interval (perhaps several minutes in duration) between the moment of impact and the formation of the central uplift,
(c) the extent of magnetic resetting decreases with increasing distance from the centre of the impact site, and
(d) the degree of resetting increases with the abundance of low coercivity magnetic grains as given by H_{cr}, a relationship found by other investigators for experimentally-produced SRM.

Intrusive breccias found on the islands yield a secondary component of TRM or TRCM origin which was frozen in after central uplift formation. Increased convergence of shatter cone axes to a central focus after rotation of host rock blocks according to palaeomagnetic data, suggests that the SRM had the same direction as that found in the breccias and hence was aligned with the ambient field at the time of impact.

Introduction

Laboratory investigations of rocks subjected to transient stresses show that a remanent magnetization can be acquired in the presence of an external field (e.g. Shapiro & Ivanov

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1966; Nagata 1971; Pohl, Bleil & Hornemann 1975; Cisowski *et al.* 1973; Cisowski & Fuller 1978). This magnetization is referred to as Pressure Remanent Magnetization (PRM) or Shock Remanent Magnetization (SRM) depending upon the time scale and magnitude of the pressure involved.

Detailed studies of the effect of shock on natural materials (Pohl *et al.* 1975; Cisowski & Fuller 1978) give results that are consistent with those of earlier weight dropping experiments (Shapiro & Ivanov 1966) and nuclear test sites (Hargraves & Perkins 1969) which suggest that the direction of the acquired magnetization is parallel to that of the external field. The hardness and intensity of SRM is also shown to increase with shock pressure (Pohl *et al.* 1975; Cisowski & Fuller 1978).

Although several palaeomagnetic investigations have been made on terrestrial meteorite impact sites with a view to detecting SRM, evidence for its existence is still fragmentary and incomplete (Hargraves & Perkins 1969; Hargraves & Roy 1974; Cisowski & Fuller 1978). The best documented examples of naturally occurring SRM are those presented by Cisowski & Fuller for the Quaternary Meteor Crater in Arizona and for the Lonar Crater in India which is less than 50 000 yr old. In both cases a critical test of SRM is to show that a secondary component of magnetization observed in rocks forming the upturned rim of the crater, aligns along a recent earths field direction after restoring the rocks to their pre-impact configuration. In the Meteor Crater study a positive fold test of this type was obtained from a limestone unit but the results are based on only three samples. In the Lonar study oriented samples for a fold test were not collected and reliance instead had to be placed upon the similarity in remanence characteristics of naturally and experimentally shocked Lonar samples.

The above studies should be distinguished from those on impact sites which have reported a remanent magnetization direction dating from the time of impact (e.g. Pohl 1965, 1971; Pohl & Soffel 1971; Robertson 1967; Larochelle & Currie 1967; Currie & Larochelle 1969; Bylund 1974). All of these studies used either melt rock or breccia samples where the remanent magnetization is thought to be of thermal or thermo-chemical origin rather than due to shock.

The purpose of this paper is to present palaeomagnetic data from the recently discovered Slate Islands impact site (Halls & Grieve 1976) and to demonstrate the existence of a Shock Remanent Magnetization in the impacted rocks. This study is of potential importance because:

(1) the occurrence of a natural SRM may ultimately permit a comparison in the estimation of palaeo-shock pressures between magnetic methods and those already established based on shock metamorphism (e.g. Robertson 1975). This aspect may be particularly important in the low pressure range (20–100 kb) where shock metamorphic zones are less secure;

(2) should SRM be a more sensitive indicator of shock it may extend the range of shock effects to lower pressures than hitherto possible, and it may also permit a greater resolution of variations in estimated shock pressure;

(3) the identification of SRM may provide a further constraint in models of crater formation; and

(4) the direction of the SRM may provide a valuable dating tool in craters lacking accessible melt rocks and breccias.

Geology of the Slate Islands

The Slate Islands comprise an approximately circular cluster of two large and 13 smaller islands in north-eastern Lake Superior (Fig. 1). They have recently been mapped by

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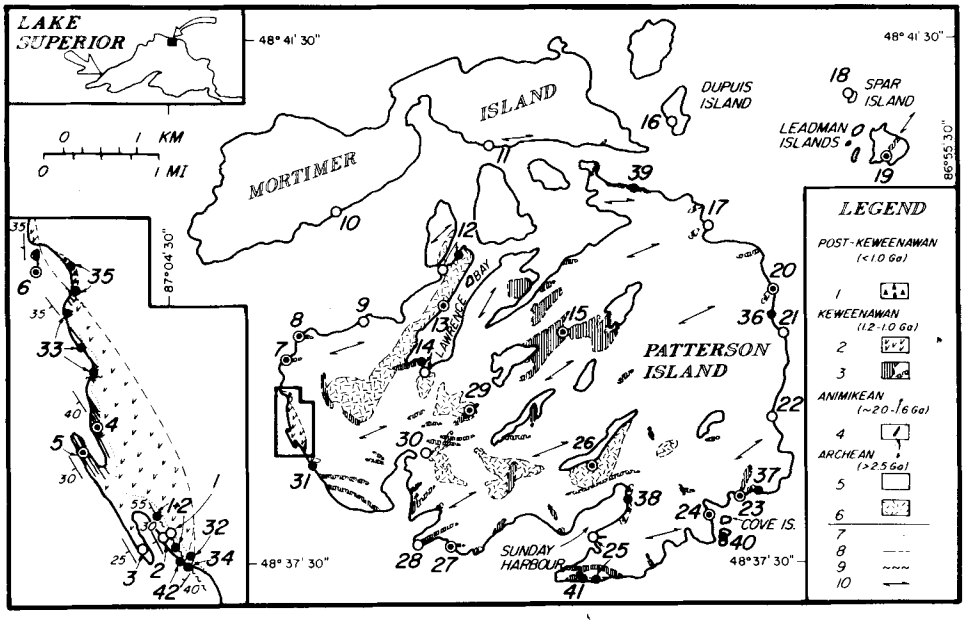


Figure 1. Geological map of the Slate Islands showing the location of shatter cone (○) and palaeomagnetic (●) sites. Sites numbered 1–30 refer to shatter cone measurement localities at which palaeomagnetic samples may or may not have been taken. Sites 31–42 are palaeomagnetic sites only. Source of geological data: Sage (1975b); inset sketch is also based in part on author's own observations. Legend: 1 – Breccia; 2 – Mafic volcanics; 3 – Diabase intrusives; 4 – Red and grey argillite overlying cherty ironstone; 5 – Predominantly mafic to felsic meta-volcanics and other igneous rocks; 6 – Meta-diorite intrusives; 7 – Limit of shoreline outcrop (inset map only); 8 – Geological contact; 9 – Fault; 10 – Foliation.

Sage (1975a, 1978) and have also been examined by the author over an accumulated period of three months between 1973 and 1977.

The islands are almost exclusively formed by metamorphosed Archean intrusive, extrusive and sedimentary rocks with a steep NE trending foliation. Along the SW coast of Patterson Island, is a sequence of mafic lava flows about 130 m thick which resembles lithologically the extensive tracts of so-called Keweenawan volcanics, about 1.0–1.2 Ga old, which border and underlie much of Lake Superior (Halls 1966). The Slate Island sequence contains about 20 flows that dip from 25 to 70° SW. The steeper dips occur at lower stratigraphic levels where much of the deformation appears to have involved movement along strike faults both within and locally at the base of the sequence (inset, Fig. 1). The volcanics have suffered only mild metamorphism in the low greenschist facies, presumably due to burial. They are underlain by a sequence of interbedded chert and ironstone about 30 m thick which is lithologically similar to the 1.8 Ga-old Animikean sequence exposed on the lakeshore about 80 km to the west (J. M. Franklin 1975, private communication). On the Slate Islands their most extensive exposure is one small outcrop at the SE end of the Keweenawan volcanic terrain (inset, Fig. 1).

Easterly to north-easterly trending diabase dykes, presumably feeders for the flows, are ubiquitous throughout the islands. They are dark grey in colour, show pronounced chilled margins and contain generally fresh pyroxene and plagioclase. The plagioclase tends to be less altered in the coarser-grained interiors. Several dykes and flows have been analysed by the A^{40}–A^{39} radiometric method, which confirms their Keweenawan age (R. Bottomley 1978, private communication).

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The youngest rocks on the islands are ramifying veins and dykes of red and green breccia which cut all other rocks including the Keweenawan volcanics and dykes. The breccias consist of angular to subrounded fragments up to 4 m along one dimension, derived locally from Precambrian rocks and set in a matrix of finely comminuted rock debris of similar origin (Sage 1978). On the east coast of Patterson Island (at site 36 in Fig. 1) a single isolated exposure of a buff breccia occurs which appears to rest on the Archean basement. The relationship of this breccia to the others is unknown.

Evidence for meteorite impact

Reconnaissance geological and palaeomagnetic work (Halls 1975) together with field and petrological studies (Grieve & Robertson 1976) suggest that the Slate Islands form the central uplift of an eroded complex meteorite impact crater about 30 km in diameter (Halls & Grieve 1976). Evidence of impact includes the occurrence of shatter cones, deformation lamellae in quartz and felspar, and a concentric pattern in lake bathymetry around the islands (Halls & Grieve 1976). Although clastic breccias are not in themselves diagnostic of impact they are commonly associated with it. The Slate Islands breccias, however, contain quartz grains with deformation lamellae and shatter-coned clasts (Halls & Grieve 1976).

Shatter cones are conical-shaped fractures, the surfaces of which carry longitudinal ridges and grooves which bifurcate away from the apex. They are considered diagnostic of intense shock (Dietz 1947) and it is found both experimentally and in the field that they point towards the shock wave source. A detailed study of shatter cones on the Slate Islands has now been completed (Stesky & Halls 1978) and shows that they are abundantly developed throughout the islands although restricted to certain rock types. The majority of cones point inwards and upwards, toward the centre of Patterson Island. The central location of the shock origin with respect to the Slate Island group, originally estimated from only four sites (Halls & Grieve 1976) is now confirmed. A summary of this new shatter cone data will be presented later with the palaeomagnetic results.

Estimated shock levels based on deformation lamellae support a centrally located shock point as they vary from about 60 kb along the SW coast of Patterson Island to about 150 kb a further 2 km inland (Grieve & Robertson 1976).

Palaeomagnetic investigations

EXPERIMENTAL PROCEDURE

Palaeomagnetic samples were obtained from 31 sites (Fig. 1), the geological descriptions of which are given in the Appendix. Owing to the rugged nature of the coastline and the often highly fractured nature of the rocks, field drilling was not undertaken and all cores instead were prepared from block samples oriented in the field using a magnetic and astro-compass. The estimated orientation error is no more than 5°. At least one core specimen from each sample was subjected to alternating field (af) cleaning using a Schonstedt SD-1 single axis demagnetizer with a maximum alternating field of 1000 Oe. All magnetic measurements were made using a DIGICO computer-assisted fluxgate magnetometer with an estimated noise level of 0.05×10^{-6} G. A description of the detailed demagnetization procedure used has been given elsewhere (Halls 1978a).

Hysteresis properties (coercive force H_c, coercivity of remanence H_{cr}, saturation magnetization J_s and saturation remanent magnetization J_{sr}) were measured for more than 60 samples using a PAR vibration magnetometer with a maximum DC field of 10 kG.

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A limited amount of thermal cleaning, on about 20 samples, was performed but was discontinued as af demagnetization was found to be a more rapid and effective means of analysing the superimposed remanence components usually found in Slate Island rocks.

MULTI-COMPONENT REMANENT MAGNETIZATION: METHODS OF SEPARATION

A reconnaissance palaeomagnetic study of the Keweenawan volcanics and dykes (Halls 1975) revealed a pervasive secondary component of magnetization superimposed on a relatively hard primary remanence. Further sites, measured from Keweenawan, Animikean and Archean rocks, have shown that the secondary component is widespread across the islands and is invariably less resistant than the earlier one to either af or thermal cleaning.

This relationship presents certain problems when trying to obtain the direction of the secondary component since it cannot be isolated from the primary. Three methods can be used under certain conditions to solve the problem:

(1) Converging circles of remagnetization; (2) vector subtraction or Zijderveld plots, and (3) stereonet plots of difference vectors. A brief description of each of these methods is given below as all of them have been used in the present study.

(1) Converging circles of remagnetization

Suppose that two components present in a rock have for reasons outlined below a difference in directional dispersion at the within sample, within site, or between site level. Providing there is merely a difference in their coercivity or blocking temperature spectra over the range of magnetic cleaning, a family or specimen, sample or site remagnetization circles will be traced out by successive total magnetization vectors, which will tend to converge about a point corresponding to the mean direction of the less-dispersed component (see Halls 1976, 1978a for further details). The direction of the convergence point can be determined by a least squares method (Halls 1976) and the required confidence limits with or without additional directional data using other methods can be obtained (Bailey & Halls 1978). The only limitation of the method is that a significant difference in component dispersion must exist. This however can be generated in a variety of ways including folding, brecciation, or conglomerate formation between the times of primary and secondary remanence acquisition (Halls 1978a). A further way is if the two components were formed over sufficiently different time intervals such that there is a difference in dispersion due to secular variation. This difference would be due to apparent polar wander if a remagnetization event had affected a number of formations whose primary components were of widely differing age.

(2) Vector subtraction and Zijderveld plots

Use of vector subtraction to obtain the direction of the less resistant component is only valid provided that over the demagnetizing interval used to obtain the successive difference vectors, the harder component remains undemagnetized (Fuller 1969; Buchan & Dunlop 1976). The only test of this condition is that the direction of successive difference vectors remains constant or at least without trend, and in this case their mean direction defines that of the softer component. This test however, fails if the ratio of primary to secondary remanence destroyed after each cleaning step remains constant. This case is intuitively a special one and the reliability of the final result may be gauged by the consistency of the estimated secondary direction within a sample population.

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A more visual representation of vector subtraction is provided by the vector diagram (Zijderveld 1967) in which the magnitude and direction of successive total vectors are projected onto two planes, one horizontal which gives changes in declination and the other vertical with an azimuth chosen ideally so as to give the least distorted view of inclination changes. If linear segments occur over the same demagnetization interval in *both* plots, the demagnetization characteristics correspond exactly with that for vector subtraction when the direction of successive difference vectors remains unchanged. As before the main restriction on the successful use of Zijderveld (Z) plots to estimate the direction of a less resistant component is that the harder component must during some measurable interval remain undemagnetized, otherwise one or both of the projections are curved and the direction of the softer component cannot be found.

(3) Stereonet plotting of difference vectors

This method is used to obtain the secondary direction whenever there is a relatively soft component such as a VRM due to the present earths field (PEF), in addition to the harder primary component. Hoffman & Day (1978) have shown that if, throughout the full coercivity/blocking temperature range of any number of superimposed components, there are defined intervals over which any two of the components only are being demagnetized simultaneously but at different rates, then the population of successive difference vectors defined in these intervals will each trace out a great circle segment lying between the directions of the two components. They show for example that, if there is a remanence of intermediate coercivity whose spectrum partially overlaps those of a high and low coercivity component, then the difference vectors obtained from these two overlapping regions will define great circles whose intersection point gives the direction of the intermediate coercivity component. In practice this method has one disadvantage: for many samples a low coercivity viscous component may constitute more than 50 per cent of the initial NRM intensity. By the time it is removed the signal to noise ratio during demagnetization of the two remaining components may be such as to obscure any great circle trend. This problem can be rectified in many instances by using difference vectors only for that part of the total spectrum where the low and intermediate components overlap, and using total vectors themselves rather than smaller difference vectors to define the higher coercivity great circle segment (Halls 1979).

In the palaeomagnetic analysis of Slate Island rocks a combined use has been made of all the above methods. The practice is to obtain total vector stereoplots, Z diagrams and difference vector stereoplots for individual specimens (Fig. 2). A complete visual description of the remanence characteristics from the viewpoint of component separation is thus obtained for a particular cleaning method.

The vector stereoplot is first used to assess whether any stable endpoints or remagnetization circles exist. For a three component case where there is a soft viscous remanence (a common occurrence due to the PEF or stray laboratory fields) two great circles may be obtained from a single specimen. If a great circle trend is not followed it is immediately known that there are three or more components. A single great circle is not of course a guarantee that only two components exist, especially in rocks where antiparallel components are found.

The Z plot is examined next and is important because it shows intensity changes and thus the amount of remanence destroyed during a particular vector trend. It will thus immediately reveal a potentially fictitious stable end-point produced merely by the inability of the cleaning method to destroy remanence. The Z plot will also show from any simultaneity of

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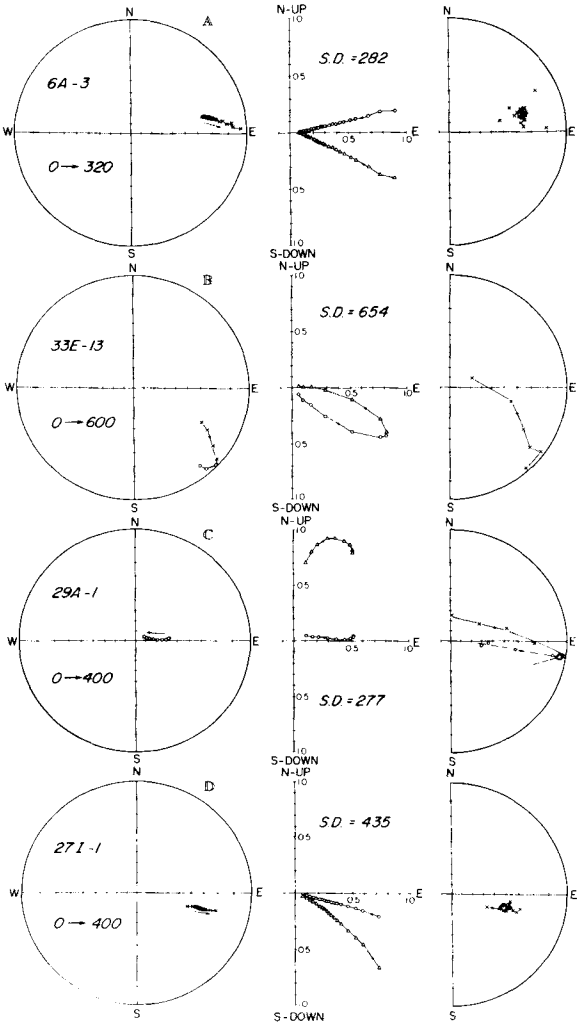


Figure 2. Vector stereoplots, Zijderveld (Z) diagrams and Hoffman–Day difference vector stereoplots for four selected Slate Island specimens. The scale division (S.D.) for Z diagrams is in magnetization intensity units of 10^{-6} G. Magnetic cleaning fields are in Oersted. Diagram A illustrates for a Keweenawan volcanic specimen, an average great circle trend but excellent linear segments on the Z plot which suggest that only the softer secondary component is being removed. The corresponding difference vectors show up as a distinct cluster on the Hoffman–Day plot. Diagram B shows another Keweenawan volcanic specimen, but one from which the direction of the secondary component cannot be obtained. Note the curved segments of the Z diagram and the corresponding Hoffman–Day plot where the difference vectors fall along a great circle close to that followed by the vectors. Diagram C shows a diabase dyke specimen where a banana-shaped path is followed by the magnetization vectors indicating that at least three components are present. The Z plot does not yield any simultaneously linear segments and yet by use of the Hoffman–Day method a component of intermediate coercivity can be obtained where the two great circle paths followed by the difference vectors intersect. Further examples of the use of the latter method are given for site 19 in Halls (1979). Diagram D shows the same behaviour as in A but for a Keweenawan diabase dyke, where an excellent separation of the secondary component direction has been obtained. Note that for all specimens above, no stable end-points corresponding to the upwardly-directed Keweenawan primary component were obtained. The maximum demagnetizing field is determined by the onset of significant scatter with accompanying destruction of the smooth demagnetization path on the vector diagram.

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linear segments whether a component can be obtained from vector subtraction. It is important here to insure that an apparently straight line segment on the inclination plot is not because of distortion due to an improperly chosen azimuth for the vertical section (an extreme example is if an E–W section is used when declinations are N–S). This problem is removed on the difference vector stereoplot because legitimate straight line segments will show up as a trendless cluster of points. The difference vector stereoplot is an essential addition because it can reveal in favourable circumstances, even when Z plots are curved, the number and direction of intermediate coercivity components (Hoffman & Day 1978). This information can only be derived from Z plots when they are composed of a series of straight line segments. The manner of presentation in the combined analysis is shown in Fig. 2 where four Slate Islands samples are used to illustrate remagnetization circles, linear and non-linear Z plots and difference vector stereoplots showing point clusters and great circle trends. Also shown is the successful use of vector subtraction (Figs 2A and D) and the Hoffman–Day method (Fig. 2C) to isolate the secondary component direction. Further examples of the application of the Hoffman–Day method to Slate Islands samples are given for site 19 in Halls (1979).

Out of the total sample population from the Slate Islands more than 70 per cent yielded clear convergent remagnetization circles (average number of data points per circle = seven; $Q < 1 \times 10^{-5}$, Halls 1978a) whereas the difference vector techniques were only able to give the secondary component direction in less than 20 per cent of the samples. A further 10 per cent carried only a single component of magnetization with a direction corresponding to that of the secondary component. It was therefore possible to gauge the reliability of the more indirect methods by comparing their results with the stable directions given by the single component samples. The majority of these samples came from breccias. A few were from country rocks either having very low H_{cr} values or situated close to the estimated shock point. The significance of these results will be discussed later.

Palaeomagnetic results

The palaeomagnetic results are summarized in Tables 1–6 which show the means by which the final primary and secondary component directions were estimated.

PRIMARY COMPONENTS

Although almost 70 samples were collected from the Keweenawan volcanics less than half yielded stable end-points, owing to the continued presence of the secondary component at the higher cleaning fields. At some sites only a single sample attained a stable end-point whereas at other more heavily collected sites as many as nine were obtained. Consequently a number of adjacent sites were combined into one so that the new sites were approximately comparable in the number of usable samples and number of flows sampled (Table 1). After correction to the palaeohorizontal the mean direction ($D = 125.9$, $I = -63$: Table 6) is less than 10° from those obtained from the Keweenawan Osler volcanics exposed on the mainland just 40 km to the west of Slate Island (Palmer 1970; Halls 1974). In the upper part of the Osler group a positive fold test indicates that this magnetization is pre-folding and was presumably formed during cooling and accompanying deuteric alteration (Halls 1974). It is therefore of probable thermochemical origin (TCRM) and may be regarded as a primary component. The similarity in direction of the component in the Slate Islands volcanics suggests that it too is of primary origin. A number of stable end-points were obtained from Keweenawan diabase dykes and in some cases their direction corresponds well, without

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Table 1. Stable end-point data for primary components. n/N is the number of specimens/samples which yielded stable end-points. S is the total number of samples measured at the site, (D_1, I_1) and (D_2, I_2) are the (declination, inclination) before and after structural correction respectively, according to the strike ($\delta°$) and dip ($\omega°$) of the layered unit. α_{95} is the radius of the 95 per cent confidence oval and k is the precision parameter.

Site	n	N	S	Number of flows	$D_1°$	$I_1°$	$\delta°$	$\omega°$	$D_2°$	$I_2°$	k	α_{95}
Keweenawan volcanics												
1 + 2	16	9	15	3	207.7	−60.3	150	50–55	90.8	−59.1	33	9.1
4	12	11	14	3	198.7	−40.6	150	35–40	155.4	−59.9	56	6.1
5	8	6	12	4	192.7	−58.4	150	30	132.8	−66.2	348	3.5
33	8	6	20	3	182.2	−49.3	140–150	40	125.9	−58.4	31	11.1
6	3	1	6	1	214.5	−82.3	178	35				
Site 31 did not yield any stable end-points corresponding to the primary direction.												
Animikean argillite-iron formation												
32	2	2	14	–	49.2	−24.0	215	40	61.0	−09.6	58	33.3
Keweenawan diabase dykes/baked host rocks												
19	6	4	14	–	55.9	−27.0	–	–	–	–	154	7.4
25	4	3	8	–	72.0	−43.0	–	–	–	–	500	3.0
39	3	2	4	–	121.1	−46.6	–	–	–	–	126	22.4
40	1	1	12	–	110.0	−74.0	–	–	–	–	–	–
41	1	1	5	–	93.0	−70.0	–	–	–	–	–	–
Sites 7, 8, 14, 15, 23, 27 and 29 did not yield any stable end-points corresponding to the primary direction.												

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need for any rotation, to the Keweenawan primary direction (sites 39–41: Table 1). At other dyke sites (19 and 25) the direction is very different (Table 1) but it will be shown that there is good evidence that these dykes have been rotated as a result of meteorite impact.

The only other component found on the Slate Islands which may be of primary origin is that found in the Animikean sequence. The only two stable samples however yield a pole position (Lat. 151° W, Long. 15° S) which is very similar to that found by Robertson (1973) in the Sibley Series, a red bed sequence, about 1.3–4 Ga old, sandwiched between the Animikean sequence and the Osler volcanics in the Thunder Bay District about 100 km west of the Slate Islands. Interestingly the two Slate Islands samples come from the uppermost 3 m of the section beneath the lowermost Keweenawan flow, where the normally grey argillite becomes red.

SECONDARY COMPONENTS

Many samples from the Slate Islands contain a soft presumably viscous magnetization due to the PEF, but the more interesting aspect is that nearly all of them contain a secondary component with easterly declination and with a downward inclination of about 45°. This component is found in the post-Keweenawan breccias and their Precambrian host rocks and it is convenient at this stage to treat these two groups of rocks separately.

Almost half of the samples collected from five breccia sites carry only a single remanence component. These stable data are summarized in Table 2 and it can be seen that the site directions agree closely. All remaining samples show small, often irregular trends on af cleaning, away from the secondary component direction, signifying the presence of another, harder component. Only those which pass the provisional Q-test (Halls 1978a) have been accepted as remagnetization circles in Table 3. The harder component is presumed to represent the net magnetization of a pre-brecciation remanence still residing in some of the clasts, because sample great circles have a random orientation (Halls 1978a, but also Fig. 3). Where sufficient data permit, the convergence point of the remagnetization circles for each site has been estimated using the least squares method of Halls (1976) together with the

Table 2. Stable end-point data for the secondary component. Symbols are as in Table 1. Only those sites are listed which did not yield any sample directions based on difference vector methods. For sites 15 and 26 the number of *specimens* was used to calculate the mean direction and associated statistics due to differences in sample weighting.

Site	n	N	S	D_1°	I_1°	k	α_{95}°
Post-Keweenawan breccias							
34	2	2	6	70.4	53.0	81	28.2
35	4	4	9	81.8	48.0	133	8.0
36	5	5	9	80.7	50.5	205	5.4
37	3	3	7	87.0	47.2	110	11.8
38	4	4	5	87.2	44.0	694	3.5
Mean site data: $N = 5$, $D = 81.8°$, $I = 48.7°$, $k = 211$, $\alpha_{95} = 5.3°$.							
Keweenawan diabase dykes							
14	7	6	6	79.8	49.6	727	2.5
15	6	4	4	126.7	21.6	69	8.1
Archean diorite intrusives							
26	6	3	3	85.4	72.2	48	9.7
Archean felsite							
20	6	5	13	91.2	53.2	230	5.1

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Table 3. Remagnetization circle data for the secondary component. Additional symbols not used in previous tables: D, I are the declination, inclination of the best intersection point (secondary direction), e_{min}, e_{max} are the 95 per cent confidence limits defined along the directions (De_{min}, Ie_{min}) and (De_{max}, Ie_{max}) respectively, obtained from the error analysis of Bailey & Halls (1978).

Site	n	N	D°	I°	$e_{min}{}^\circ$	$e_{max}{}^\circ$	$De_{min}{}^\circ$	$Ie_{min}{}^\circ$	$De_{max}{}^\circ$	$Ie_{max}{}^\circ$
Keweenawan volcanics										
1?	10	5	87.0	21.8	4.1	17	182.4	33.6	220.5	−48.2
2?	14	8	81.5	50.0	5.5	15	200.0	21.8	124.3	−31.7
4	13	12	58.2	51.6	1.9	8.6	240.8	38.4	329.8	−01.3
5	15	11	49.0	59.7	3.5	37	208.6	28.7	303.5	8.9
33	29	19	99.3	36.4	4.1	30	219.3	34.2	158.0	−35.2
6	11	6	73.4	27.2	6.0	25	346.2	−09.6	275.5	62.8
31	5	5	97.1	40.9	4.1	16	207.5	22.0	138.1	−41.1
Post-Keweenawan breccias										
34	3	3	77.4	54.5	34	48	350.0	−01.9	261.3	35.4
35	6	5	83.8	55.7	6.2	6.4	329.1	15.9	249.8	29.5
37	5	5	76.9	49.3	10	13	357.8	−09.3	275.5	39.2
Keweenawan diabase dykes										
7	10	5	103.5	48.7	8.2	26	218.2	20.2	142.7	−34.3
19	21	11	147.6	58.6	4.4	32	335.6	31.2	63.4	−03.5
25	15	8	104.6	59.5	4.7	66	352.7	12.4	76.2	−27.4
27	11	8	90.5	27.9	4.7	25	000.5	00.1	90.2	−62.1
29	5	3	87.3	−15.0	26	180	180.7	−12.5	129.0	70.3
39	5	4	104.8	51.4	7.1	22	10.9	03.1	98.4	−38.4
40	8	8	353.0	79.2	4.7	31	187.7	10.4	277.2	−02.7
41	5	5	272.6	71.4	9.6	110	356.7	−02.0	86.0	18.5
42	4	4	60.1	63.3	12	40	206.8	22.8	302.4	13.2
Animikean argillite-iron formation										
32	16	12	80.3	40.6	5.3	28	329.4	22.6	38.2	−40.9
Archean felsite										
20	6	6	83.0	55.3	7.1	16	109.7	15.6	296.1	30.1

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Table 4. Combined stable end-point and difference vector data for the secondary component. N is the number of samples, n_{sep} the number of specimens providing stable end-points, n_{vs} the number of specimens yielding the secondary component direction by vector subtraction (Fuller 1969), and n_{mhd} is the number of specimens yielding the direction by the Hoffman & Day (1978) method as modified by Halls (1979).

Site	n_{sep}	n_{vs}	n_{mhd}	N	$D°$	$I°$	k	$\alpha_{95}°$
Keweenawan volcanics								
2?	1	5	0	4	86.9	47.4	140	7.8
6	0	10	0	5	73.4	27.2	184	5.7
Keweenawan diabase dykes								
7	3	5	0	5	96.0	53.1	77	8.8
19	0	1	10	5	127.6	55.0	84	8.4
25	0	4	1	4	104.9	54.5	175	7.0
27	0	7	0	4	89.4	39.6	180	6.9
Archean diorite intrusives								
12	3	3	0	3	100.8	46.2	107	12.0
13	4	2	0	3	93.5	43.3	1567	3.1
Animikean argillite-iron formation								
32	1	6	0	5	88.4	41.0	52	10.7

Sites 4, 5, 31 and 33 in Keweenawan volcanics did not yield the secondary component direction on the basis of SEP, VS or MHD techniques.

error analysis of Bailey & Halls (1978). Results shown in Table 3 agree well with those derived using the stable directions given in Table 2. If the results are now combined for each site using all stable samples and only those *additional* samples which yield remagnetization circles (Bailey & Halls 1978) the final estimate of the secondary component direction is obtained for each breccia site (Table 5). The best estimate of the secondary component direction for reasons that emerge later is given by the mean value of these breccia sites ($D = 81.9°$, $I = +49.7°$, Table 6). The close agreement of magnetization directions between breccia sites on the Slate Islands has been found before at other impact craters (e.g. Pohl 1965; Pohl & Soffel 1971). The occurrence in Slate Islands breccia of possible altered clasts of diaplectic minerals and rock glass (Halls & Grieve 1976) suggest elevated temperatures and hence in accordance with the findings of Pohl (1971) on other impact breccias the remanence is thought to be of TRM or TCRM origin. Rock magnetic studies reported later are generally in accord with this conclusion.

The Keweenawan, Animikean and Archean rocks into which the breccias have been injected have been analysed in a slightly different way. Samples exhibiting remagnetization circles are even more prevalent and well-defined, but due to the shortage of stable end-points, difference vector methods were used wherever possible to rectify the deficiency in directional estimates from individual specimens.

As before Table 3 summarizes the remagnetization circle data for each site where possible and Table 4 gives combined results from stable end-points, vector subtraction and the modified Hoffman–Day method. Only four sites relied only on stable end-points and these are given in Table 2. Table 5 presents the final site estimates based on all independent pieces of data.

A surprising and curious result emerges from a comparison of Tables 3 and 4. For the majority of sites convergent magnetization circles are observed at the *within* site level and furthermore yield estimates of the secondary component direction which agree well with

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Table 5. Final estimates of site secondary component directions. N_{sd} is the number of samples in which the secondary component direction has been found from either stable end-points or difference vector methods. N_{rc} is the total number of *additional* samples yielding remagnetization circles ($\leqslant N$ in Table 3). Of eight samples measured at site 27 only those oriented by sun compass are included. Directions and statistics for all sites yielding both specimen directions and remagnetization circles have been calculated using the method of Bailey & Halls (1978).

Site	N_{sd}	N_{rc}	D°	I°	$e_{min}{}^\circ$	$e_{max}{}^\circ$	$\alpha_{95}{}^\circ$
1?	0	5	87.0	21.8	4.1	17.0	
2?	4	5	86.9	47.4	4.5	6.0	
4	0	12	58.2	51.6	1.9	8.6	
5	0	11	49.0	59.7	3.5	37	
6	5	0	73.4	27.2			5.7
7	5	0	96.0	53.1			8.8
12	3	0	100.8	46.2			12.0
13	3	0	93.5	43.3			3.1
14	6	0	79.8	49.6			2.5
15	4	0	126.7	21.6			8.1
19	5	6	127.1	56.1	4.5	6.5	
20	5	6	88.1	53.1	3.5	4.5	
25	4	4	102.8	55.1	4.3	6.0	
26	3	0	85.4	72.2			9.7
27	4	0	89.4	39.6			6.9
29	1	3	87.3	−15.0	26	180	
31	0	5	97.1	40.9	4.1	16	
32	5	6	76.7	36.4	5.7	8.2	
33	0	19	99.3	36.4	4.1	30	
34	2	3	75.0	53.9	6.3	7.7	
35	4	5	82.4	51.0	5.3	5.4	
36	5	0	80.7	50.5			5.4
37	3	3	82.8	48.8	7.3	8.5	
38	4	0	87.2	44.0			3.5
39	1	4	103.8	50.3	3.3	6.0	
40	0	8	353.0	79.2	4.7	31	
41	0	5	272.6	71.4	9.6	110	
42	0	4	60.1	63.3	12	40	

those obtained by other methods. This result can only be explained if the secondary component has a dispersion which is less than that of the associated characteristic remanence. Extensive studies on Keweenawan volcanics and dykes elsewhere around Lake Superior (Palmer 1970; Halls 1974; Pesonen 1978) suggest that the characteristic component is a primary TRM or TCRM. On the Slate Islands many of the sites are either from Keweenawan dykes less than 10 m thick (Nos 19, 25 and 39) or from two to three adjacent flows (Nos 4–6) or a single flow (No. 31) so the primary was formed during cooling in perhaps less than 10 yr. The low extent of hydrothermal alteration in either dykes or flows precludes the possibility that the characteristic component is a CRM which was slowly acquired during burial. Thus the characteristic component was formed rapidly and consequently its within-site dispersion should be small. The fact that it is not implies that either (a) it was somehow dispersed prior to or during the formation of the secondary, or (b) more preferably for reasons discussed later (p. 583) the secondary component originally had an unusually low dispersion, much less than the primary, but that due to joint-controlled deformation accompanying crater formation the dispersion of *both* was increased.

A summary of all sample data for the secondary component is given in Fig. 3. Many of the data points are averages of two or more specimens. A few data points in addition to those

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Table 6. Palaeomagnetic pole positions (reversed) derived from primary and secondary components. N is the number of sites; dm, dp are the semi-axes of the 95 per cent confidence oval about the pole position. Corresponding data for the Animikean rocks are not given as the pole is based on only two samples.

Sampling locality: Slate Islands, northern Lake Superior, Lat. 48.67° N, Long. 86.93° W

Remanence	Age	N	$\bar{D}$	$\bar{I}$	k	α_{95}°	Lat.	Long.	dP	dM
Primary	Keweenawan (~ 1.1 Ga).	4	125.9°	−63.0°	37	15.4°	53.3° N	162.6° W	19.0°	24.2°
	Data obtained from Keweenawan volcanic sites (1 + 2), 4, 5 and (6 + 33) after correction to the palaeohorizontal, using the observed attitude of flows.									
Secondary	Post-Keweenawan (0.5–0.9 Ga. ?)	5	81.9°	+49.7°	307°	4.4°	27.5° S	167.1° E	3.9°	5.8°
	Data obtained from Post-Keweenawan breccia sites 34–38 inclusive, with the assumption that no tectonic rotation has occurred after the breccias cooled following their injection during impact.									

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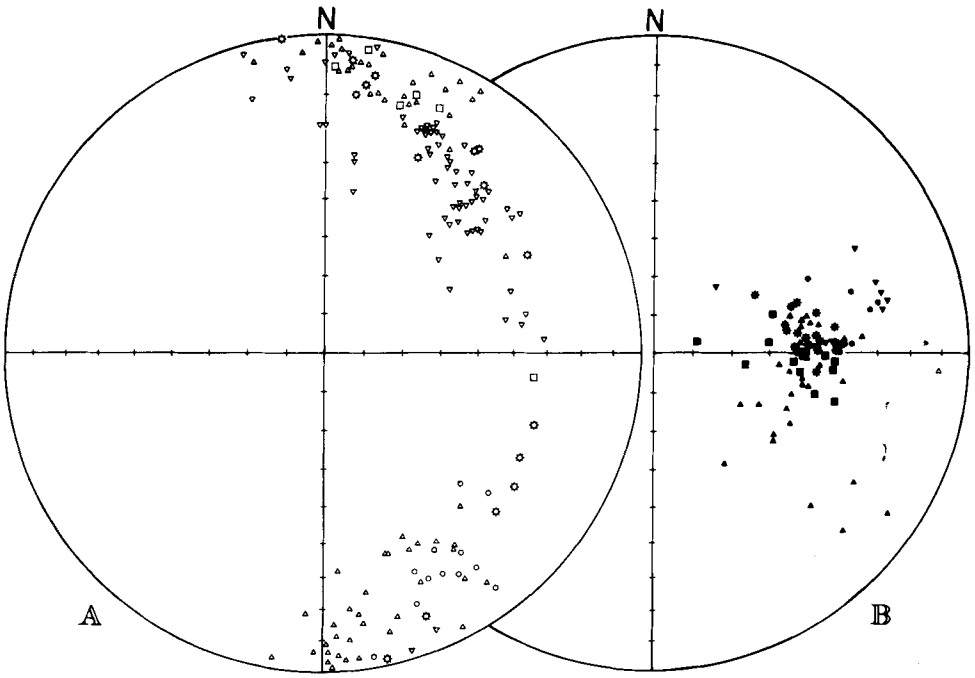


Figure 3. Equal area projections showing A: distribution of poles to sample remagnetization circles and B: distribution of sample secondary component directions from stable end-points, vector subtraction and the modified difference vector method (Hoffman & Day 1978; Halls 1979). Symbols used: ◦ Post-Keweenawan breccias; ▽ Keweenawan volcanics; △ Keweenawan dykes; ○ Animikean sedimentary rocks and □ Archean rocks. Open/closed symbols refer to upward/downward pointing directions.

represented in Table 3 have been included and come from two sparsely sampled dykes from the south and south-west shores of Patterson Island. In Fig. 3A poles to sample remagnetization circles have been plotted. The normal to the best fitting great circle drawn through this girdle of points would lie close to the centre of the cluster shown in Fig. 3B which is formed by sample directions derived from stable end-point, vector subtraction and modified Hoffman–Day methods. Less than 50 per cent of the data points in Fig. 3B are obtained from the samples used to compile Fig. 3A. It should be noted that the density of points along the great circle girdle in Fig. 3A is very uneven. Fig. 4 is a simplified sketch which shows that this unevenness arises from the restricted azimuth of sample great circles caused mainly by the limited azimuthal spread of primary remanence directions. The only data points which occupy the thinly populated segment in Fig. 3A between declinations 90 and 140° are derived from the breccias. Note that the breccia points *are* evenly distributed along the girdle reflecting the randomly directed component due to the clasts which is present in addition to the secondary component.

It is tempting at this stage of the analysis, considering the overall consistency of the data to take the grand mean of all site directions as the best estimate of the secondary component direction. However, this is tantamount to assuming that there has been no differential rotation between sites after the component was acquired. That such rotations have occurred is suggested by Fig. 3B where directions derived from the breccias are more tightly grouped than the corresponding data from their enclosing host rocks. This same observation also holds at the site level where in Table 5 and Fig. 5 the closely grouped directions from breccia sites 34–38 can be compared with the more dispersed host rocks results. Where the type and

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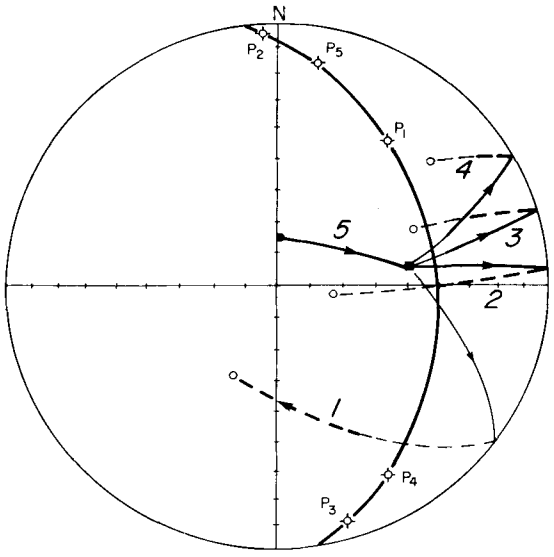


Figure 4. Simplified explanation for the distribution of points in Fig. 3A. Great circles 1–5 are each representative of major host rock groups and show the typical paths that the remanence vector follows on demagnetization, (the thickened portion shows the average path length defined by data points). For each remagnetization circle the secondary component direction ■, and that of the other components (○, ●) characteristic of that rock group are shown. 1 – Keweenawan volcanics; 2 and 3 – Keweenawan diabase dykes; 4 – Animikean sedimentary rocks; 5 – Archean rocks. For great circles 1–4 the secondary component is less resistant to cleaning than the other one which is an earlier remanence of probable primary origin. For great circle 5 the other component is softer than the secondary and may be a recent viscous remanence. P_1–P_5 represent poles to the great circles and their distribution is similar to that in Fig. 3A. The pole tb the thick-lined great circle drawn through these points defines the secondary component direction.

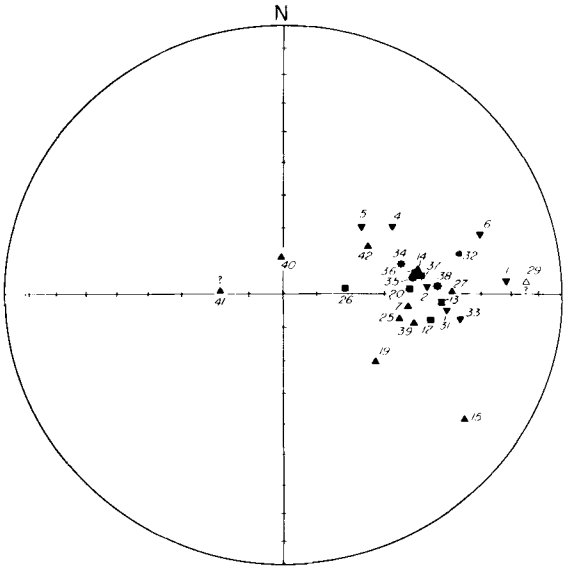


Figure 5. Equal area stereonet summarizing all *site* secondary component directions according to sampled rock type. Symbols are the same as in Fig. 3. Directions from sites 29, 33 and 40–42 are uncertain and are excluded from further analysis. The site 29 direction may be real on the basis of shatter cone measurements (see Appendix).

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quantity of data afford comparison the within-site scatter for both breccia and host rocks are very similar. Some of the large uncertainties in direction at many host rock sites is because a total reliance had to be made on converging remagnetization circles with limited azimuthal spread. Therefore the difference between breccia and host rock dispersion at the between-site level cannot be explained by any within-site cause.

In earlier papers (Halls 1975, 1978a) it was suggested that the relatively small dispersion of the secondary component compared to the primary in Keweenawan rocks was due to the secondary having been acquired in a time scale short compared to that of secular variation. The occurrence of this component together with shatter cones, both unique features in Keweenawan rocks (Halls 1975) suggests that the remanence was acquired presumably instantaneously in the 2–3 s during passage of a shock wave following meteorite impact. The difference between breccia and host rock dispersion at the between site level can be easily reconciled with meteorite impact using the following model.

An intense shock wave with pressure in excess of 50 kb travels outward from the impact point, forming shatter cones and imparting an SRM to the country rocks parallel to the ambient external field. A transient cavity about 7 km in radius is excavated (Dence in Grieve & Robertson 1976) with injection of breccia into pre-existing joints and newly-formed cracks. This stage is followed by central uplift formation due to inward displacement of the crater rim and/or upward movement of the crater floor. The mechanics of crater formation suggest that the creation of the central uplift would be initiated several seconds after impact (Milton & Roddy 1972) and would be essentially complete after several minutes (M. J. Dence & P. B. Robertson 1978, private communications). The ponding of melt rocks around the central uplift of the Mistastin Crater, Labrador (Grieve 1975) for example testifies to its rapid formation. The breccia meanwhile having been heated through friction during transport and perhaps by energy transfer from the impacting body, begins to cool gradually. It thus acquires a partial TRM or TCRM in the direction of the ambient field which is frozen in *after* the host rocks (and their SRM) have been rotated to form the central uplift.

COMBINED PALAEOMAGNETIC AND SHATTER CONE ANALYSIS

If the above sequence of events is correct, the tectonically disturbed host rock directions, when rotated into that given by the breccias, may result in an increased convergence of shatter cone apices toward a common focus.

A number of sites where both palaeomagnetic and shatter cone measurements are available were thus subject to further analyses. Twelve of these sites for which geological and palaeomagnetic data are relatively simple are shown in Fig. 6. Palaeomagnetic data from two geologically more complex areas are given and discussed in Fig. 7.

In theory it is possible to estimate the true orientation of the rotation axis for any site given directional estimates for primary and secondary component both before and after deformation. Unfortunately for the only sites where this condition is approached (Nos 4–6), there is considerable uncertainty in the primary direction prior to deformation. The reason is that the volcanics may have had a more southerly dip prior to impact in conformity with the general structure of the Lake Superior basin (Halls 1966), and have been rotated into their present attitude during formation of the central uplift. A single stage structural correction to obtain the true primary direction at sites 4–6 may thus be invalid. Although the mean direction for the primary component after correction is close to that expected on the basis of nearby data from the mainland Osler volcanics, there may be considerable error, perhaps up to 20°, at individual sites. An alternative procedure, which could also be used at diabase

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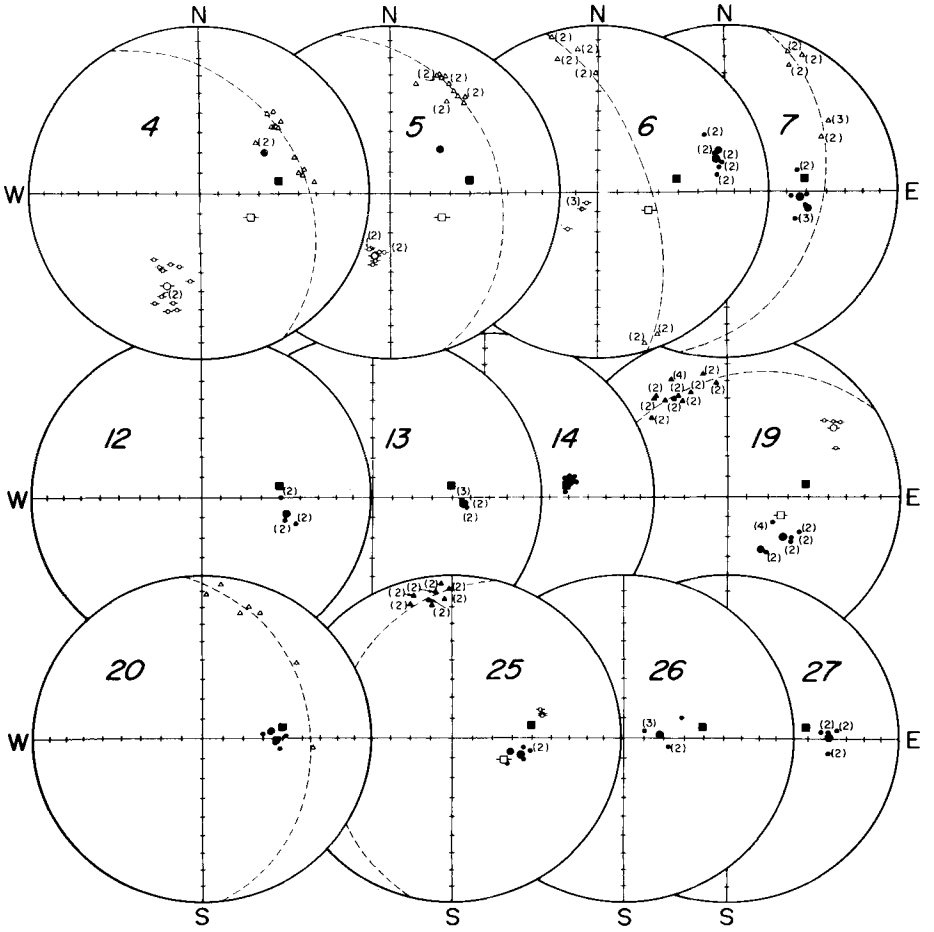


Figure 6. Equal area stereonets summarizing palaeomagnetic data for 12 Slate Islands sites. Symbols used: -○- sample primary component direction given by stable end-point; -0- mean of sample primary directions; -□- primary component direction, after structural correction, for the Keweenawan Osler Volcanics (mean of results obtained by Palmer 1970 and Halls 1974); ■ secondary component direction given by mean of five breccia sites; • secondary component direction in host rock sample given by stable end-point or vector difference methods; ● mean direction of secondary component sample data; △, ▲ normal to sample remagnetization circle; ◔ normal to great circle (dashed line) fitted by least squares to sample normals. Solid/open symbols are downward/upward pointing directions. Sample points may themselves be means of specimen data. The number of specimens used, if more than one, is given beside points.

dyke sites 19 and 25, would be to assume that the primary direction is the same as that given by the Osler volcanics (Palmer 1970; Halls 1974). However, the between-site dispersion found for those rocks is such that departure of up to 20° from this direction may be anticipated for any one site.

In view of these uncertainties and also because primary direction information is lacking at many sites, all rotations have been performed about assumed horizontal axes. The azimuth of these axes and the degree and sense of required rotation are shown for the various sites in Figs 8 and 9. The orientation of shatter cone axes and other structural elements (pole to bedding or dike plane) are each shown before and after rotation. That there is a significant difference between breccia and host rock directions at most sites can be seen in Fig. 10

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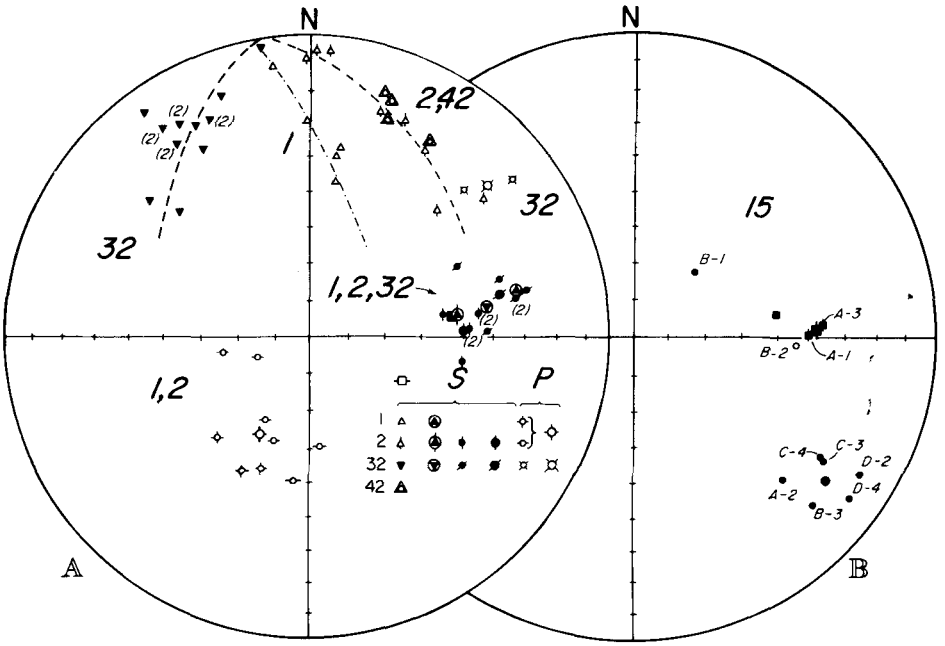


Figure 7. Equal area stereonets summarizing palaeomagnetic data from two areas of complex geology. Data in diagram A are taken from a small area at the southern end of the inset map in Fig. 1 which includes four sites: two in Keweenawan volcanics (1, 2), one in a diabase dike (42) and one in Animikean sedimentary rocks (32). The Keweenawan volcanics have an irregular dip of 30–55° SW, with higher dips occurring at lower stratigraphic levels. A strike fault occurs toward the base of the sequence (Sage 1975b). Shatter cone measurements at two sites (1 and 2) on either side of this fault show an abrupt increase in negative axial plunge from 16° at site 2 to 32° at site 1 where the flows are 20° steeper. Palaeomagnetic samples were taken two years prior to the shatter cone study but their position with respect to the fault (unrecognized at that time) was not recorded. Although dips of flows from which samples were collected range from only 50 to 55°, it is surmised that samples were taken from either side of the fault and thus from the general region of both sites 1 and 2 because the poles to sample remagnetization circles form two distinct girdles. The normal to one girdle yields a secondary direction virtually identical to that given by the breccias, whereas the other yields a shallower secondary direction. If steepening of shatter cone axes occurred after secondary component acquisition as results from other sites show, samples yielding shallow secondary inclinations would come from site 1 and the others from site 2. Remagnetization circle, vector subtraction and stable end-point data from the immediately underlying Animikean strata also yield a shallow secondary direction. Symbols for each site are shown in diagram A and from first column onwards correspond in order to: (1) pole to sample remagnetization circle, (2) normal to great circle through sample poles, (3) secondary component direction from vector subtraction and/or stable end-points, (4) mean of sample directions, (5) sample primary component direction and, (6) mean of sample primary directions. Other symbols are the same as in Fig. 5. In diagram B four samples from site 15 were obtained from the interior and margin of a diabase body intruded into heavily shatter-coned Archean rocks. Two samples, C and D, carry only a single component. Sample D was from the coarse-grained interior, the other from the margin. Sample A, also from the margin, was found to be a composite of diabase and breccia. Specimen A-2 composed of diabase gave a stable direction close to that of the other samples, while specimens A-1 and A-3 which were dominantly formed of breccia gave stable directions within about 10° of that given by the five breccia sites 34–38 (■). Sample B, also from the margin, appeared to be made up of diabase fragments and gave highly scattered but stable specimen directions, one of which was similar to those given by samples C and D. Symbols are the same as given in Fig. 6.

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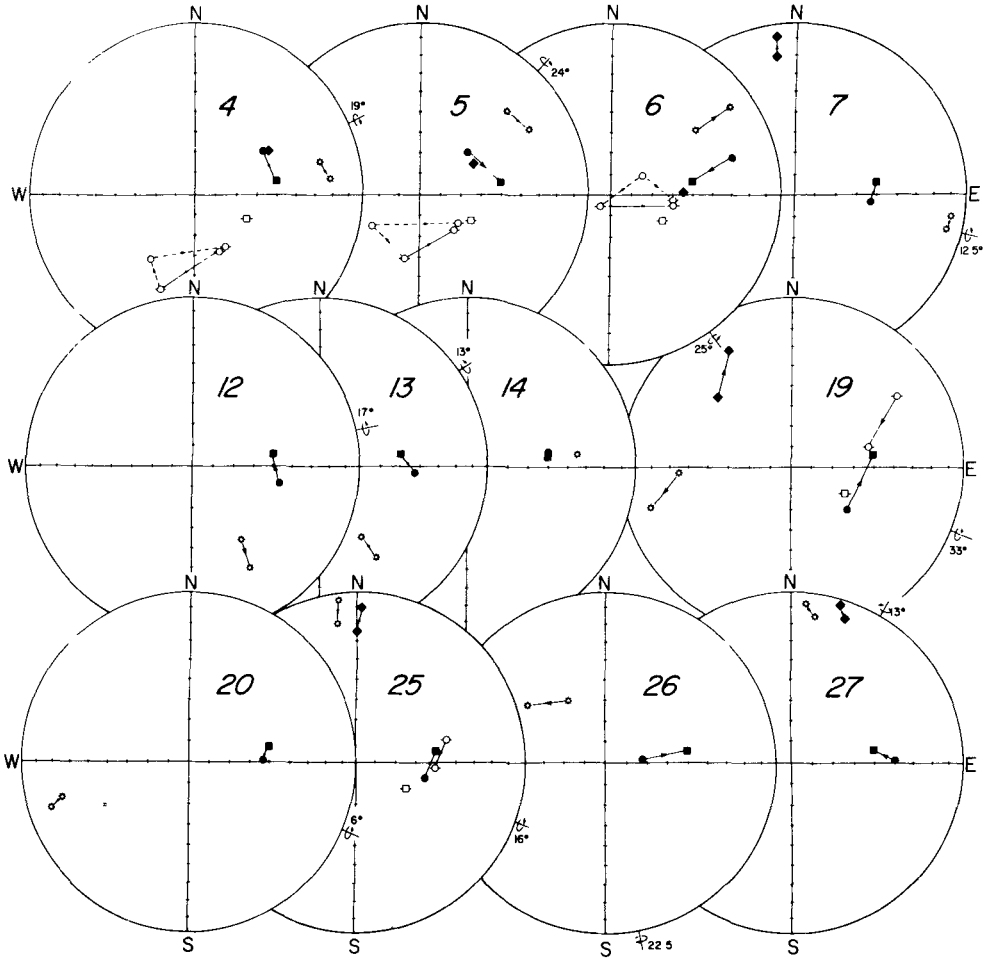


Figure 8. Equal area stereonets showing horizontal axis rotation of secondary remanence direction given by host rocks sites (Table 5) to that given by the breccias (Table 5) for the 12 sites in Fig. 6. The corresponding directional movement of the primary component (-o-), shatter cone axes (⌘), and normal to bedding or dyke plane (♦) is also shown. Other symbols are the same as in Fig. 6. At sites 4–6 alternative movement paths are shown for the primary component: that based on correction to the palaeohorizontal (pole moves to vertical) using observed dip of flows (→–) and a two-stage rotation based first on the secondary component data followed by correction to paleohorizontal (– → – –). There is little difference between either method in the final primary direction obtained.

where 95 per cent confidence ovals are shown. These observations are important because if the breccia and host rock components were originally acquired along the same external field direction, then there is evidence on palaeomagnetic grounds alone that significant rotations have taken place.

A summary of the shatter cone orientation data is shown in Fig. 11A and is based on measurements at 30 sites (Stesky & Halls 1978). Although cones generally point inwards and upwards toward the centre of the island group, there are a number of sites where they point in anomalous directions. Since these discrepancies in many cases are much greater than measurement error, the divergence of cones away from the central focus is thought to be caused by rotation of allochthonous blocks during central uplift formation. In Fig. 11B only those sites are shown for which palaeomagnetic data have already been presented in

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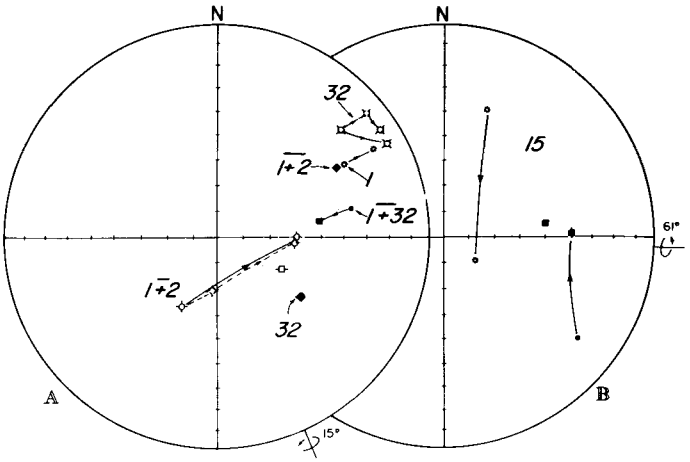


Figure 9. Equal area stereonets showing the horizontal axis rotation as explained in Fig. 8 for sites whose palaeomagnetic data were shown in Fig. 7. Symbols the same as in Fig. 8. Data for some sites (e.g. 1 and 32) have been combined. Rotation of the secondary component (•) and shatter cone directions (☆) are around the axis shown, whereas primary directions (✧ and ⌑) have been rotated in the same way as in Fig. 8.

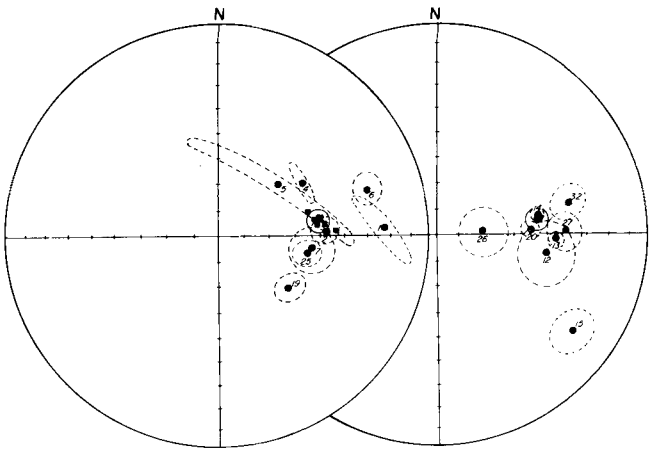


Figure 10. Equal area stereonets summarizing site directions for secondary component at shatter-coned sites. Breccia (▪) and host rock (•) sites and the mean breccia site direction (■) are distinguished. Using Tables 5 and 6, 95 per cent confidence ovals are drawn for all host rock sites (dashed line) and also for the mean breccia site direction (solid line). Results show that many host rock sites yield secondary component directions which are significantly different at the 95 per cent confidence level from that given by the breccias.

Figs 6 and 7. In Fig. 11C the same sites are shown but with the shatter cones rotated according to the palaeomagnetic data. The shatter cone and rotation data are summarized in Table 7 which also gives the uncertainty in the position of the shock focus both before and after the performed rotations. This calculation, made by R. M. Stesky, shows that there is a significant improvement in shatter cone convergence after rotation. In this analysis only 11 of the 15 sites listed in Table 8 were used. Sites 1 and 2 were excluded because the exact relation between the location of palaeomagnetic samples and shatter cone measurements was not considered sufficiently well documented, and site 5 was rejected because the declination of the secondary component is ill-defined (Fig. 6). Site 15 was also not included

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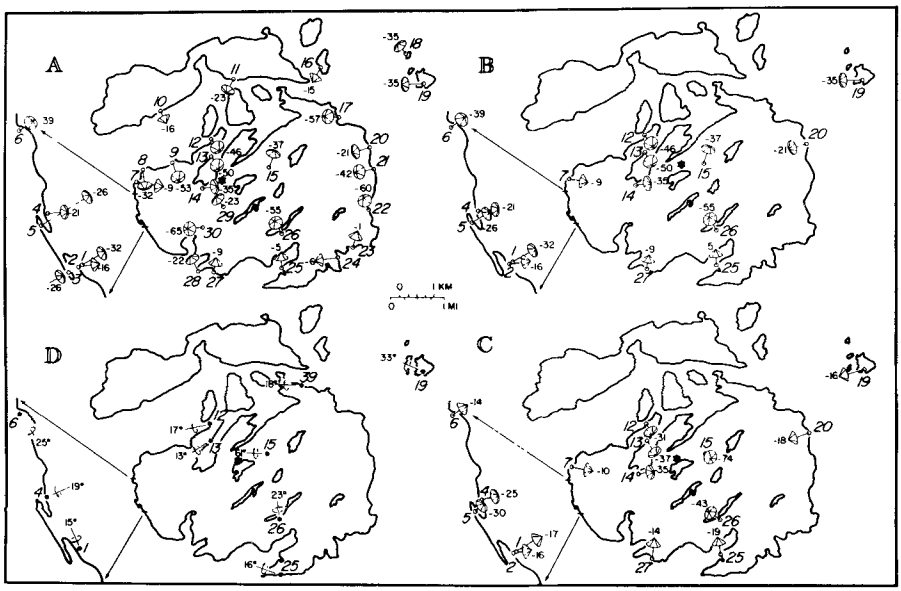


Figure 11. Shatter cone orientation measurements from the Slate Islands (data from Stesky & Halls 1978, and in preparation). Small numbers beside schematic shatter cone representations give the upward or negative inclination of cone axes. The black star gives the approximate location of the impact point as deduced from the shatter cone information. In diagram A the location is obtained using all sites and in diagrams B and C by using only the 11 most reliable ones (see Table 7). Diagram A shows all shatter cone measurements and in B only those for which palaeomagnetic data exist. Diagram C shows the same cones as in B but rotated according to the palaeomagnetic results. The increased convergence of cones toward a central focus is due primarily to an increased conformity in inclination of sites 6, 19, 20, 25 and 27, situated about 4–6 km from focus, and also to a significant change in azimuth shown by cones at sites 15 and 19. Diagram D shows the azimuth of axes from the largest rotations. The sense of movement has been reversed to approximate the original tectonic disturbance. Note that the sense of movement, particularly along the north and western shorelines of Patterson Island, corresponds to increased uplift toward the centre of the island and that rotation axes are approximately tangential around this uplift.

but only because of difficulty in incorporating an overturned shatter cone direction into the least squares calculation, where the sum of squared deviations from the convergence point to the axial projections of cones was minimized. The main cluster of stable end-point directions for site 15 in Fig. 7B is considered to represent a complete resetting of the diabase by the secondary magnetization event for two reasons: (1) on the rotation to the breccia direction there is a dramatic movement of shatter cones to a steep upward direction in keeping with the close proximity of the site to the projected impact point (Figs 11B and C) and (2) a complete resetting of the remanence is expected insofar that the chilled margin of the vertical diabase dyke at site 14, a comparable distance from the impact point, is totally reset (Fig. 6).

The considerable improvement in shatter cone convergence after rotation suggests that dispersion in cone directions is more a function of rotation than of inward translation of allochthonous blocks. This result does not necessarily indicate that inward movement was minimal, only that it contributed less to shatter cone dispersion. However, it is possible with increasing depth in a central uplift that movement becomes dominated by rotation rather than by inward telescoping, a mode of deformation that may be more characteristic of shallower depths. In view of the postulated early Palaeozoic to Late Precambrian age of the Slate Islands impact (p. 583) the present central uplift may be deeply eroded and thus

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Table 7. Shatter cone orientation data and least squares analysis of shock point position. (T_1, P_1) and (T_2, P_2) are the (trend, plunge) of the shatter cone axis (measured in the direction of cone apex) obtained respectively before and after rotation according to palaeomagnetic data. θ° and ϕ° are the azimuth of horizontal rotation axis and the amount of rotation respectively. The notation corresponds to the right-hand screw rule (thumb in the direction of axis, direction of fingers indicates sense of rotation).

Site	T_1°	P_1°	θ°	ϕ°	T_2	P_2
1	61	−32	335	15	61	−17★
2	81	−16	0	0	81	−16★
4	84	−21	246	19	76	−25
5	59	−26	223	24	47	−30★
6	54	−39	324	25	54	−14
7	100	−09	04	13	102	−10
12	145	−46	77	17	150	−31
13	149	−50	56	13	148	−37
14	83	−35	0	0	83	−35
15	18	−37	92	61	126	−74★
19	268	−35	112	33	254	−16
20	254	−21	113	06	252	−18
25	353	−05	110	16	352	−19
26	329	−56	168	23	306	−43
27	06	−09	32	13	09	−14

Least squares analysis of shock point location

All sites used except those marked by ●

	Elevation above present land surface (km)	Uncertainty in horizontal position (km)
Before rotation	1.4 ± 0.6	± 0.6 E–W; ± 0.6 N–S
After rotation	1.1 ± 0.3	± 0.4 E–W; ± 0.3 N–S

lateral movement may indeed be negligible. The improved shatter cone convergence in Fig. 11C also indicates that rotation axes were close to horizontal perhaps within 20°, a result that is physically reasonable in view of the manner in which the central uplift is thought to have formed.

In Fig. 11D the rotation axes are shown for all palaeomagnetic sites where there appears to be a significant difference between breccia and host rock secondary directions. The sense of rotation given in Table 7 has been reversed to approximate the original tectonic motion. It is noteworthy that the majority of axes are tangential about the central impact point, and that the sense of movement is generally consistent with that arising from central uplift formation. Apart from the scatter in shatter cone directions there is little direct geological evidence for the relative block movements revealed by the palaeomagnetic data. Sage (1978) reports only minimal differential rotation between country rock blocks across breccia zones, but this observation is not surprising if the breccia was injected prior to most of the deformation. Sage also comments that the breccias have suffered little tectonism themselves, but again in terms of meteorite impact, signs of tectonism would be minimal if the hot breccia was still fluid during central uplift formation.

A more positive indication of tectonism is that diabase dykes (at sites 7, 19, 25 and 27) which are generally intruded parallel to host rock foliation, assume a more vertical attitude after performed rotations (Fig. 8). This effect is particularly conspicuous at site 19 on the easterly Leadman Islands where the trend of the dike and enclosing host rock foliation becomes more ENE, similar to that of the regional grain on the main islands to the west. In addition the primary magnetization at sites 19 and 26 moves on rotation toward the direction expected for Keweenawan igneous rocks.

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Table 8. Hysteresis and remanence intensity data for site 19 from the Leadman Islands. H is the range of cleaning fields over which a clear great circle path is followed by the resultant vector, N_s is the number of cleaning steps in this interval, J^p_{200} and J^s_{200} are the intensities of primary and secondary component respectively after a cleaning field of 200 Oe, H_c is coercive force, H_{cr} coercivity of remanence, J_s saturation magnetization, J_{sr} saturation remanence, T_c is Curie temperature (from J_s–T curves obtained on a magnetic pendulum balance in the Geophysics Institute, Bergen, Norway) and d is the distance of each sample from the margin of the diabase dyke (about 10 m wide). All values of hysteresis parameters represent the mean of two powdered samples taken from opposite ends of each specimen. *e*: Values of J^p_{200}, J^s_{200} and F (defined in Fig. 13) were obtained by backward extrapolation of the component demagnetization plots, assuming exponential decay. For values of cleaning field below H a third component due to the PEF was present so that it is in this interval that extrapolation was made. The assumption of exponentiality appears reasonable for the secondary component. It is likely invalid for the primary but usually the primary decay was comparatively gradual so that the exact form of the curve used in extrapolation was not important. Polished sections of each diabase specimen suggest that opaque oxides have been exposed to little post-deuteric oxidation. Except for differences in grain size (smaller close to the chilled margin) there is no major difference in opaque mineralogy: Titanomagnetites in deuteric oxidation class I–III show up to 100 per cent granulation in smaller grains. Ilmenite both free and intergrown/exsolved with titanomagnetite is unaltered. The granulation features are anisotropic, patchy in occurrence, non-magnetic, and are probably rutile/anatase. There is no evidence of hematite or iron hydroxides.

SPECIMEN	N_s	H	J^p_{200}	J^s_{200}	$J^p_{200}+J^s_{200}$	F	H_{cr}	H_{cr}/H_c	J_{sr}	J_{sr}/J_s	T_c	d
		(Oe)	($G \times 10^{-6}$)			(%)	(Oe)		(Gcm^3/g)		(^oC)	(cm)
F-3	8	50-400	1335	557	1892	29	347	1.7	0.18	0.25	585	~ 5
B-1	6	100-500	1142	734	1876	39	360	1.8	0.28	0.24	595	~ 5
C-3	7	100-250	948	396	1344	29	307	1.9	0.25	0.20	590	~ 5
D-3	6	100-500	649	539	1188	45	282	2.1	0.24	0.16	585	20
A-3	7	125-275	379	726	1105	66	265	2.2	0.27	0.15	590	30
E-1	12	75-350	209	275	484	57	255	4.7	0.16	0.09	575	45
K-1	6	125-200	190	338	528	64	245	3.2	0.22	0.10	565	> 60
M-1	15	200-330	327	283	610	46	325	2.2	0.16	0.17	550	> 60
M-2	15	200-380	276	256	532	48	272	2.1	0.16	0.19	550	> 60
N-2	8	175-350	235	272	507	54	237	3.1	0.14	0.14	540	> 60
R-1	5	275-400	295e	510e	805	63	227	4.9	0.18	0.18	550	> 60

Certainly deformation on the Slate Islands has occurred subsequent to Keweenawan igneous activity as witnessed by the tilted and faulted volcanics, and by faulted and twisted dykes at sites 7 and 8 respectively. How much of this deformation is due to normal tectonic causes accompanying the formation of the Laker Superior basin and the larger Keweenawan rift systems (Halls 1966, 1978b) is unknown. It should be noted however that the maximum 60–70° dips observed in the volcanics is by far the greatest observed along the northern limb of the Lake Superior basin, where volcanics normally dip about 10–40° lakeward, so that an additional disturbing influence such as meteorite impact is conceivable. Two very small outcrops of friable quartzose sandstone occur as possible fault slices on the west coast of Dupuis Island (Fig. 1). These rocks lithologically resemble the late Keweenawan Bayfield–Jacobsville sandstone which on seismic evidence (Halls & West 1971) occurs as a flat-lying blanket which floors most of the Lake Superior including that part in the vicinity of the Slate Islands. At least some of the deformation on the Slate Islands appears therefore to postdate Keweenawan time, although this is not necessarily anomalous within the tectonic framework of the Lake Superior basin as major boundary faults along both north and south limbs of the structure were active after Bayfield–Jacobsville time. The only clues that an additional factor, unrelated to Keweenawan basin development, has occurred on the Slate Islands are the shock deformation features related to a central, local focus and the highly anomalous direction of the secondary component, quite unlike any of those found in Keweenawan rocks.

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In summary three lines of evidence suggest that rotations performed on the basis of palaeomagnetic data reflect deformation occurring during the formation of the central uplift: (1) there is a marked increase in convergence of shatter cone axes to a common focus, (2) the azimuth of rotation axes and the sense of rotation about them appears consistent with that expected during central uplift formation, and (3) at diabase dyke sites, differences in geological structure and anomalies in primary magnetization direction are reduced after rotation.

CORRELATION BETWEEN PALAEOMAGNETIC RESULTS AND HYSTERESIS PROPERTIES

The palaeomagnetic results presented so far appear to be consistent with the secondary component having been produced by shock. If indeed an SRM has been formed in the rocks of the Slate Islands it might be expected that the extent of magnetic resetting decreases away from the shock point and furthermore using experimental shock results of Cisowski & Fuller (1978) there should be an inverse relationship between the degree of magnetic resetting and the value of sample coercivity measured either by H_c, the magnetic coercive force, or by H_{cr}, the remanence coercivity.

The first hint that there is some dependence upon coercivity is found in Keweenawan diabase dykes, where coarse-grained interiors appear to be more reset than finer-grained chill margins. This effect is seen to some extent in all diabase dykes examined but is particularly well documented at sites such as 19 (Table 8), 25 and 39. A striking example is shown in Fig. 12 from site 39 where chilled margin and baked host rock samples remain relatively unaffected compared to coarse-grained, more interior samples. This site also provides a good example of remagnetization circles converging to the less-dispersed secondary direction.

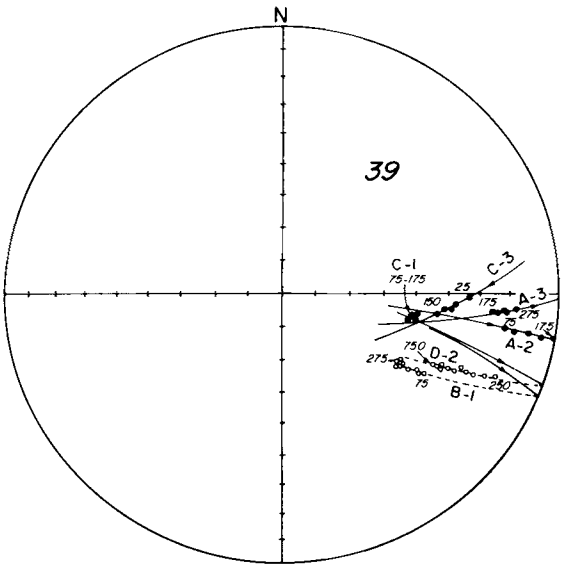


Figure 12. Behaviour of four samples from diabase dyke site 39, on af cleaning. The dyke thins eastwards from about 3 to 0.2 m thick. Samples A and B are from chilled margins of the 3 and 0.2 m thick dykes respectively. C is 0.6 m from margin and D is a baked host rock adjacent to chilled margin. The range of cleaning fields shown corresponds to the interval between removal of low coercivity PEF component and the appearance of significant noise. The remanence surviving in these intervals, expressed as a percent of initial NRM intensity, is for each specimen as follows: C-1, 60→15; C-3, 90→30; A-2, 65→35; A-3, 30→10; B-1, 80→20; and D-2, 106→80.

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Estimates of both primary and secondary component directions are available for many samples taken from Keweenawan volcanics and dykes. Given both directions of a two component system, it is easy to resolve the observed resultant vector along these directions for each cleaning field. Thus for the cleaning interval over which a great circle path is traced out by successive resultant vectors, intensity demagnetization plots can be obtained for both components. Hence the ratio of secondary to primary magnetization intensity remaining after any given cleaning field may be found. A slightly different quantity F is used here and is defined as the secondary intensity J_s expressed as a percentage of the total scalar intensity $(J_s + J_p)$ after a cleaning field of 200 Oe. A 200 Oe field value was chosen because by then more than 90 per cent of the samples follow remagnetization circles after removal of a third viscous component. It is thus a field which is common to most of the different cleaning intervals over which samples exhibit two-component behaviour. For some samples the primary direction could only be obtained from the point of closest approach of the great circle to the mean direction for that site. This procedure is of course incorrect but it provides a consistent method for obtaining the primary direction for these samples. It may lead to errors of up to perhaps 10° measured along the great circle but these do not change the calculated F values by more than about 10 per cent. This is because the angle between the primary and secondary directions is about 100–120° and most 200 Oe data points lie closer to the secondary rather than to the primary direction.

Measurements of hysteresis properties (H_c, H_{cr}, J_s and J_{sr}) were carried out on all samples from which primary and secondary demagnetization curves could be obtained. Samples were then divided according to their horizontal distance from the vertical line through the shock point so that magnetic properties of samples having experienced approximately the same shock pressure could be studied. The samples could be grouped conveniently into

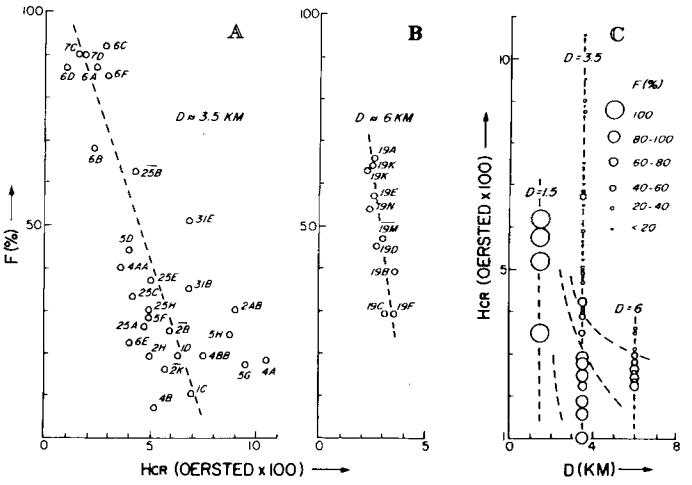


Figure 13. Diagrams A and B show plots of F versus remanent coercivity H_{cr} for two sample populations, where D, the horizontal distance from the vertical line through the shock point is 3.5 and 6 km respectively. $F(\%) = J^s_{200}/(J^p_{200} + J^s_{200}) \times 100$ where J^s_{200} and J^p_{200} are respectively the intensities of the secondary and primary components after a cleaning field of 200 Oe. F is thus the secondary component intensity expressed as a percentage of the scalar sum of primary and secondary intensities. The scalar rather than the vectorial sum is used because it is independent of the directions of the components and also because it gives a crude measure of the total number of aligned domains forming the two components. H_{cr} rather than other hysteresis parameters such as H_c was used because it gives the bulk coercivity of all potential NRM carriers and may thus be more closely related to F. Diagram C shows values of F as a function of both H_{cr} and distance D.

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three populations having distances of 6 km (site 19), 3.5 km (sites 1, 2, 4–7, 25 and 31) and 1.5 km (site 14). It was found for both the 3.5 and 6 km populations that if H_{cr} is plotted against F an inverse relation is obtained (Figs 13A, B).

Figs 13A and B show, in agreement with the findings of Cisowski & Fuller (1978) that samples with low H_{cr} are more easily reset by a given shock pressure than those with high H_{cr}. Also noteworthy is the well-defined linear relationship at diabase dyke site 19 (Fig. 13B and Table 8), inviting speculation that the slope of this line may give a measure of the shock pressure and would presumably increase to zero with increasing pressure as all grains ultimately become reset. The graph in Fig. 13A for rocks closer to the impact point does indeed appear to have a higher slope but the data are scattered. This larger scatter may be attributable to a variety of causes in concert, two of which are: (1) the graph is a composite of a number of sites which due to possible differences in the amount of inward translation may not have experienced the same shock pressure and (2) deuteric oxidation of magnetite in the volcanics has occurred, particularly in flow tops, with the formation of hematite. Some of the very high H_{cr} values in these rocks are undoubtedly caused by hematite as sample 4A, for example, yields high blocking temperatures between 600 and 700°C on thermal demagnetization (see Fig. 16C) and also the highest H_{cr} value (over 1000 Oe). The presence of hematite may be especially serious as it may not respond in the same way as magnetite to shock (Cisowski 1977).

To examine the linear relationship further and to test whether it gives a measure of shock pressure, further experiments, including laboratory shock wave studies, on Slate Islands rocks are obviously required.

A comparison of the extent of magnetic resetting at the three distances 1.5, 3.6 and 6 km from the centre of the impact structure is shown in Fig. 13C. It is apparent that for samples having approximately the same H_{cr} values, the degree of resetting increases toward the impact point.

The results of this section in summary suggest:

(1) that the amount of magnetic overprinting is spatially related to the centre of the Slate Islands, the projected region of impact, and appears to decrease with increasing distance from it, and

(2) there is a dependence of secondary component acquisition on H_{cr} which is similar to that found by Cisowski & Fuller (1978) for experimentally produced SRM.

ARM–TRM studies

A preliminary set of ARM–TRM experiments was carried out on breccia and host rock samples in an effort to see whether there was any difference in the character of their secondary component. Only those samples were used which showed a high degree of directional stability throughout the full range of either af or thermal demagnetization. In several of the breccia samples a very small swing (from 2–5°) to lower inclinations was revealed, particularly on thermal cleaning during removal of the first 60–70 per cent of the remanence. The origin of this trend is unknown but does not appear to be the same as the much longer trends of different azimuth shown by most breccia samples on af cleaning. The experimental procedure used in the ARM–TRM studies is as follows.

Samples, after af demagnetization of their NRM to 1000 Oe, were given an anhysteretic remanence (ARM_1) along the direction of the previously erased component ($H = 0.5$ Oe; $\tilde{H} = 1000$ Oe). This remanence was then af demagnetized to 1000 Oe and a total TRM imparted to the rock, again along the secondary component direction. As before, the external field was 0.5 Oe and the maximum heating temperature was that which on the basis of

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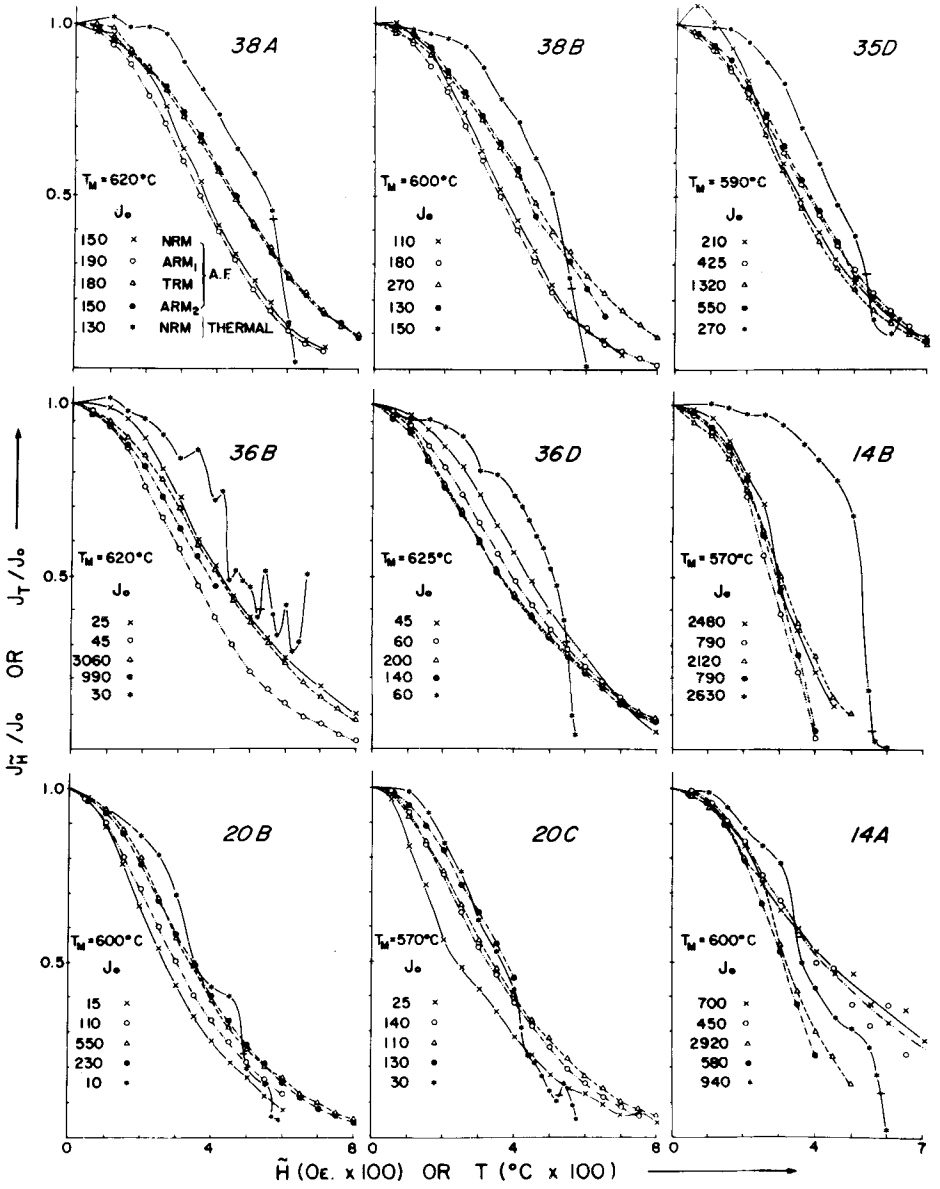


Figure 14. Combined sample af/thermal demagnetization plots of different remanence types from breccia sites 35, 36 and 38 and from host rock sites 14 and 20. All af curves are a function only of remanence coercivities lying between zero and 1000 Oe. Any remanence surviving a 1000 Oe cleaning has been vectorally subtracted from all intensities obtained at lower cleaning fields. T_m is the maximum heating temperature for the artificial TRM, $\tilde{H}$ is the alternating field intensity in Oe, T is the heating temperature in °C, J_x is the remanent intensity remaining after a value of $\tilde{H}$ or T equal to x, J_0 the initial intensity is in units of $G \times 10^{-6}$. The TRM and ARMs were all produced in dc fields ranging from 0.55 to 0.58 Oe, and the ARMs in an ac field of 1000 Oe.

previous thermal cleaning of a companion specimen was just sufficient to remove more than 95 per cent of any remanence surviving from the prior ARM demagnetization. The artificial TRM was then af demagnetized to 1000 Oe and a further anhysteretic remanence (ARM_2) was given to the rock with the same field direction and settings as for the ARM_1. The ARM_2

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was then also demagnetized to 1000 Oe. A comparison between the decay curves and intensity of the ARMs thus permitted assessment of the extent of physicochemical change occurring as a result of the intervening TRM application. Although results are preliminary (Fig. 14), as further tests together with polished section analysis are needed, a few observations pertinent to the problem can be made.

BRECCIAS

Two types of remanence characteristics are shown by the breccias:

(1) At site 38 the intensity of the TRM is larger but comparable (less than three times) to that of the NRM. The ARM_1 and NRM decay curves are very similar but there appears to be conversion of grains to higher coercivities after heating as the intensity of ARM_2 is less than that of ARM_1 but it has a slightly greater hardness. The similarity of ARM_2 to the artificial TRM suggests that for these specimens ARM is a satisfactory model of TRM. The similarity of ARM_1 to the NRM decay curve therefore suggests but does not necessarily demonstrate that the NRM is a TRM. Compared to other breccia sites (see below) the TRM intensity is only slightly larger than that of the NRM suggesting that most clasts have been magnetically reset. It is possible that randomly oriented clasts carrying remanence in hematite are present as several reddish siltstone clasts occur in this breccia (Halls & Grieve 1976). These would not be reset during artificial TRM acquisition as the maximum temperature used was 600–620°C. Site 38 is the only one of the five breccia sites where remagnetization circle trends on af cleaning are comparatively rare (only one specimen from five samples). The above data are thus compatible with the NRM being a PTRM which has succeeded in resetting the majority of grains.

(2) Samples from two other breccia sites (35 and 36) differ from those above in four ways:

(a) the TRM intensity is more than five times that of the NRM;
(b) there is an increase in the number/volume of grains susceptible to ARM acquisition after heating;
(c) ARM does not appear to be such a good model of the TRM and
(d) the bulk of the coherent remanence is destroyed at lower temperature (less than 550°C).

At site 36 the NRM is also distinctly harder than ARM_1 and there is evidence of severe chemical change in specimen 36B. At site 35 the large increase of TRM intensity compared to NRM does not appear to be caused by physicochemical change as the ARM_1 and ARM_2 plots are very similar. Despite the intense red colour of the rocks at this site and thus the suspicion of hematite, the thermal demagnetization plot shows that the bulk of the coherent NRM is destroyed at temperatures well below the Curie temperature of pure hematite (675°C).

The above results appear to be consistent with the secondary component being a low temperature PTRM at site 35. Sage (1978) reports that breccia clasts in general show little evidence of alteration around their margins and that the breccia dykes themselves have had only a minor effect on adjacent host rock to the extent of low temperature hydrothermal-type changes. It is possible of course that temperatures were higher but that chemically active fluids were absent. Perhaps noteworthy is that of the three breccia sites studied, the only one having experienced a more complete magnetic resetting of the grains (No. 38) is also closest to the impact point.

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HOST ROCKS

Two host rocks sites (14 and 20), which appear to contain *only* the secondary component were studied. At site 14 (the diabase dyke close to the impact point) two samples gave different results. Specimen 14B showed no physicochemical change on heating; the TRM had an intensity comparable to the NRM, and the NRM, ARM and TRM decay curves were all very similar. These observations accord well with a TRM origin for the NRM. The second sample 14A showed evidence of physicochemical change on heating as the ARM_1 and ARM_2 intensity and decay curves are not the same. The ARM intensity is moreover much stronger than the NRM. A viscously-decayed TRM origin might be suspected for the NRM especially since a considerable fraction of low Curie temperature grains are shown by the thermal decay curve. Alternatively it is possible in view of the dissimilar nature of the ARM plots that additional low coercivity material has been created by heating and has led to the increase in artificial TRM intensity.

NRM characteristics of site 20 (the pink felsite of probable Archean age) appear quite different from any of the previous sites described above. Both the intensity and hardness of the NRM are much less than either ARM or TRM. Physicochemical changes have occurred through heating in sample 20B but not in sample 20C. In sample 20C therefore only a small fraction of potential remanence carriers contributes to the observed remanence. This component is unlikely to be a viscously-decayed TRM as this might discriminate against lower coercivity grains and yield a harder remanence than ARM or TRM, contrary to the observations in samples 20B, C.

If the above host rocks sites have acquired their remanence by shock then the very different behaviour shown by sites 14 and 20 may be a consequence of their different distances from the impact point. At site 14 for example where estimated shock pressures are about 130 kb (Grieve & Robertson 1976) it is possible that an SRM assumes TRM characteristics because at such elevated pressures adiabatic heating under compression may become important. However, experimental results on quartz and feldspar (see Stöffler 1971) suggest that peak temperatures at such pressures may not exceed about 150°C. If the remanence at site 14 *is* a TRM, it may have been acquired through heating from above by high temperature fall-back breccia or melt, although there is no evidence of the expected resetting of the A^{40}–A^{39} age (p. 583).

In summary the ARM–TRM results are not definitive as regards the TRM or TCRM of the breccias as opposed to the SRM carried by the host rocks but suggest that depending upon the shock pressure SRM acquisition may involve interplay between pressure and temperature effects, the latter predominating increasingly toward the shock point.

Discussion

EVIDENCE FOR SHOCK REMANENT MAGNETIZATION

Evidence that the secondary component in the impacted rocks of the Slate Islands is a Shock Remanent Magnetization may be summarized as follows:

(1) Keweenawan rocks which are shatter-coned carry the secondary component whereas those remote from the Slate Islands in other parts of Lake Superior neither exhibit shatter cones nor the secondary component.

(2) If the observed difference in secondary component direction between breccia and host rocks is removed by rotation about horizontal axes an increased degree of convergence of shatter cones to a central focus is obtained.

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(3) The axial orientation and sense of rotation is broadly consistent with the manner of deformation expected in central uplift formation.

(4) There appears to be a decrease in the amount of magnetic overprinting with increasing distance from the impact point.

(5) The dispersion of the secondary component is less than that of the primary even at sites where the primary component dispersion should be small.

(6) There is an inverse relationship between the extent of magnetic resetting along the secondary component direction and H_{cr}, the remanent coercive force, which is similar to that reported by Cisowski & Fuller (1978) for experimentally-produced SRM.

The first four observations are all linked directly in some way with products of meteorite impact, namely the shatter cones or the central uplift, and suggest therefore that the secondary component is genetically linked both temporally and spatially with these features. Observations (2) and (5) suggest that the secondary component was rapidly formed at least in the country rocks. The estimated time scale of central uplift formation, perhaps a few minutes, indicates that the remanence was formed virtually instantaneously which of course would immediately separate it from TRMs and CRMs formed in more conventional geological ways. This instantaneous formation is not however reflected in the within-site dispersion where directions obtained from sample populations show dispersion similar to or even greater than remanences acquired in normal geological processes. This dispersion may be inflated due to specimen orientation errors, although the very tight grouping shown by sites 13, 14 and 38 indicates that the block sampling method is capable of high precision. The comparatively large dispersion at many sites such as 19, 25 and 32, where many samples were obtained, suggests that there is some other cause for the observed dispersion. Either the remanence was acquired over a time interval comparable to that of secular variation cycles which would immediately invalidate an SRM interpretation, or that the dispersion has been increased after remanence acquisition. The latter explanation is favoured because at the within site level the remagnetization circles indicate a lower secondary dispersion compared to that of a primary component which at many sites should have been acquired in a geologically short period of time (p. 565). Site 25 for example is from a diabase dyke only 3 m thick which would presumably have acquired its primary remanence in one or two years, and yet this component has a significantly greater dispersion than the secondary. Possible explanations are (a) that it has been somehow randomized during the remagnetization process or (b) that small differential movements along joint planes and other fractures have occurred. The second explanation is preferred not only because such movements might be expected during central uplift formation and would increase the dispersion in *both* primary and secondary components, but also because as Figs 6–7 and Table 7 show, sites with the lowest dispersion tend to have suffered the least rotation.

A further indication that the secondary component was rapidly formed is that A^{40}–A^{39} heating experiments repeatedly give ages of 1.0–0.9 Ga for the Keweenawan volcanics and dykes (Bottomley & York, unpublished data). These ages are slightly lower than the 1.1 Ga value normally assigned Keweenawan magmatism and yet they are similar to the many K–Ar ages that have been obtained from Keweenawan igneous rocks around Lake Superior. There is little indication, even in the most magnetically reset dyke (at site 14), where shock pressures are estimated by Grieve & Robertson (1976) to have been about 130 kb of a disturbed or partially reset argon content. If indeed the ages at this site are totally reset then the secondary magnetization and presumably the shock event associated with it are more than 900 Ma old. However, Hartung, Dence & Adams (1971) have reported that K–Ar ages are little affected by shock pressures less than about 140 kb, so that it is simpler to

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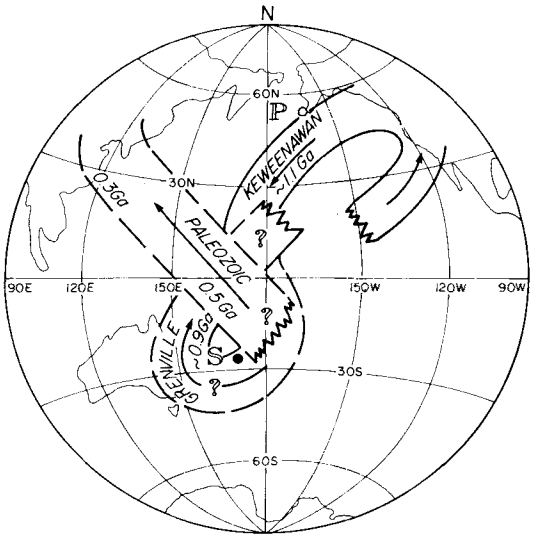


Figure 15. Apparent polar wander path for North America (after Irving, Emslie & Ueno 1974). The poles corresponding to the Keweenawan primary component (P) and the secondary component (S) are marked. The S pole position lies in a complex part of the polar path where there are trends defined from Keweenawan, Grenville and Palaeozoic rocks. The age of the secondary magnetization on the Slate Islands can thus only be placed within a broad time period of about 900–500 Ma.

assume that the Slate Islands ages more closely reflect the time of Keweenawan igneous activity and that the remanence is an SRM. Unfortunately the apparent polar wander path from North America can only define the age of the secondary component within very broad limits. The secondary pole position falls in a region of the path where its age could lie anywhere from 0.9 to 0.5 Ga (Fig. 15).

One of the more difficult problems is how to experimentally distinguish an SRM from other more usual types such as TRM or CRM. The relationship between secondary component acquisition and H_{cr} suggest some mechanism which preferentially resets low coercivity grains. This relation has been demonstrated for an SRM (Cisowski & Fuller 1978) but is not necessarily to be expected for either a TRM or PTRM where high coercivity grains could be activated nor for a CRM where the size distribution of newly-formed grains carrying the secondary component should not necessarily be confined to low coercivity types.

Most of the palaeomagnetic sites on the Slate Islands lie at distances greater than 3 km from the impact point where estimated shock pressures are less than 100 kb (Grieve & Robertson 1976). Therefore unless the rocks were highly porous it would be unlikely that residual or even peak temperatures would rise much above 150°C (Stöffler 1971). If however, the degree of magnetic resetting is more a function of temperature than pressure (i.e. a TRM) it might be expected that thermal rather than magnetic cleaning may be the most efficient way of removing the component. Since the component appears to have formed rapidly, the temperature–time relations of magnetic blocking (Pullaiah *et al.* 1975) would suggest that the temperature at which the remanence is finally removed may give an estimate of the maximum temperature reached and this may in turn show a decrease away from the shock point. Accordingly, a number of volcanic and diabase samples were thermally cleaned. The results (Fig. 16) were however disappointing insofar that for any temperature below 600°C a clear stable end-point corresponding to the primary direction (and

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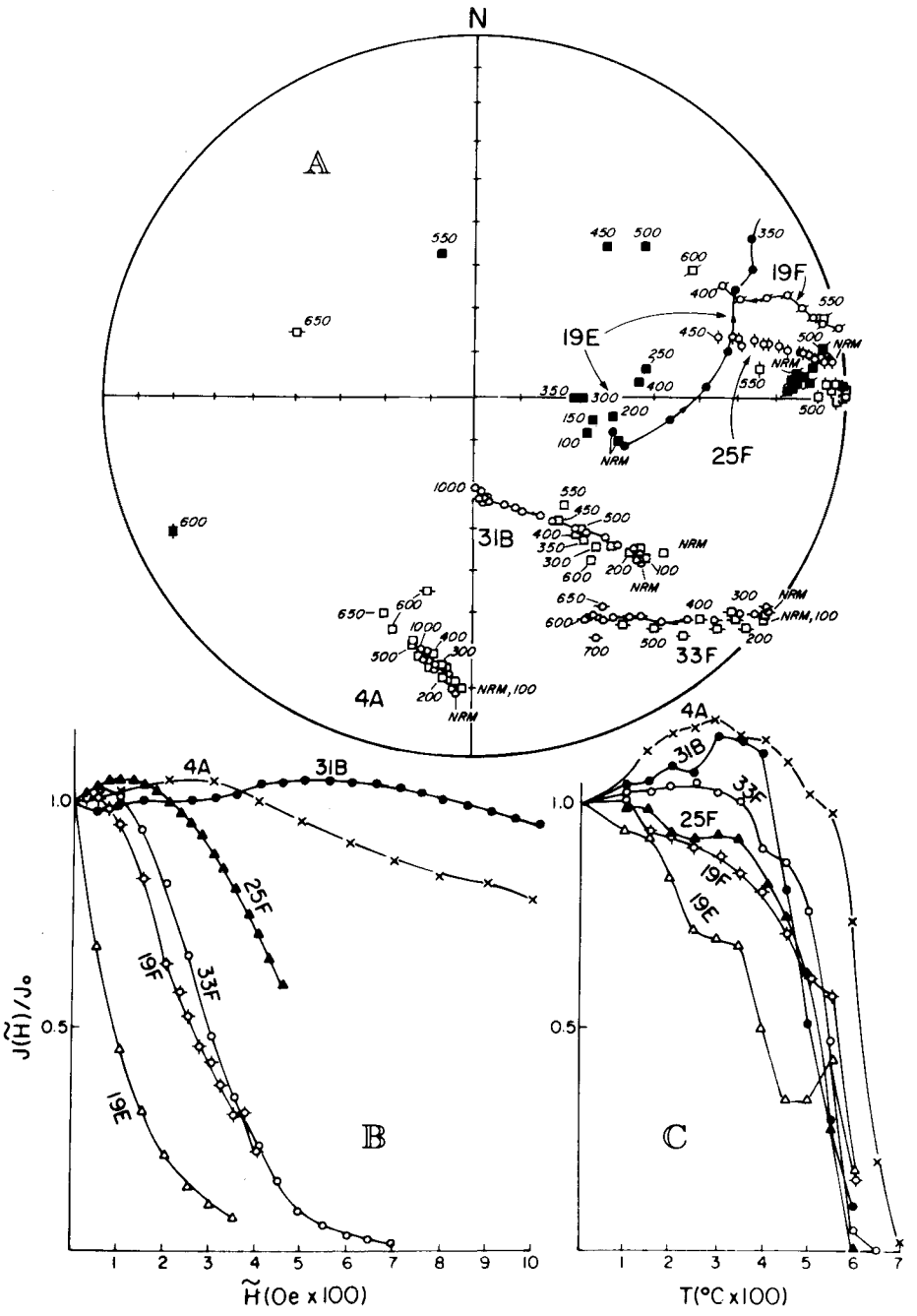


Figure 16. A comparison between af and thermal cleaning results for selected Slate Islands specimens. Diagram A; equal area stereoplot of magnetization directions (round/square symbols: af/thermal). Diagrams B and C are respectively the corresponding specimen af and thermal decay curves.

thus implying total removal of the secondary) was not obtained. Although the secondary was preferentially destroyed, both components persisted at temperatures between 500 and 600°C, by which time the total intensity had dropped to less than 10 per cent of its initial value and directions became scattered. A comparison of af and thermal cleaning for selected

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samples (Fig. 16) shows that in most cases thermal cleaning was a less effective means for component separation (e.g. samples 19E and 31B). The thermal results therefore suggest that the secondary component is not a rapidly acquired PTRM. They are in fact more compatible with an SRM origin as Cisowski & Fuller (1978) have shown that experimentally produced SRM carried by low coercivity grains and thus sensitive to af cleaning, can be relatively difficult to erase thermally as it may activate grains spanning the full range of blocking temperatures.

FURTHER IMPLICATIONS

If an SRM interpretation is accepted, the rotations shown in Figs 8 and 9 imply that the SRM component was formed essentially parallel to the ambient field in support of findings by previous workers. A number of studies indicate that stress-induced magnetization may be deflected from the ambient field either towards or away from the maximum stress direction (see Cisowski 1977, pp. 135–137 for summary). It is conceivable that some of the scatter in secondary component direction shown in Fig. 5 may be caused by this factor. However, it is certainly not a dominant cause otherwise at each site in Figs 8 and 9 the maximum stress direction (which may be given by the shatter cone axis), the ambient field direction (as given by the breccias), and the observed secondary direction, would lie along great circles which they are not observed to do.

A further point concerns the possible use of magnetic methods in the estimation of palaeo-shock pressures. Laboratory investigations by Cisowski & Fuller (1978) and others suggest that the amount of SRM increases with shock pressure and that progressively higher coercivity fractions are affected as the pressure increases. It should therefore be possible to express the degree of SRM acquisition in terms of the coercivity characteristics of the rock and the applied pressure. In this way a magnetic calibration scale similar to that based on shock metamorphism (e.g. Robertson 1975; Stöffler 1971) could be established. However, before such a scheme is useful it is necessary to identify naturally-occurring SRM so that a comparison of palaeo-shock pressures using both the magnetic and metamorphic deformation methods can be made. Although palaeomagnetic investigations of country rocks in the vicinity of impact sites have been made, none with the possible exception of Meteor and Lonar craters have demonstrated the existence of an SRM. This is perhaps because the majority of rocks measured (Triassic red beds, Grenville anorthosites) have had very high H_{cr} values and thus have been magnetically insensitive to shock. Future studies of SRM in the field should thus concentrate on low coercivity material.

One of the limitations of the use of shock metamorphic features (Robertson 1975; Grieve & Robertson 1976) is that the development of planar features in quartz and felspar may be dependent not only upon the peak shock pressure but also on its time duration which in turn is determined by the width and propagation speed of the shock front. The appearance of certain diagnostic lamellar orientations in minerals may thus occur in natural conditions at lower pressure than indicated by laboratory experiments where shorter time scales prevail. It is conceivable, although undemonstrated, that magnetism may be less sensitive to the time duration of the shock front than the inelastic behaviour of minerals and may thus provide a more reliable way to estimate palaeo-shock pressures. Some time dependence has however already been shown by Pohl *et al.* (1975) where several concussions were needed on rock samples before a limiting value of SRM was reached. The short time of compression (about 10^{-5} s) in these experiments may not have allowed sufficient time for complete response of domain movements to the applied stress.

Some final observations concern possible uses of SRM in models of crater formation

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In areas such as the Slate Islands where pre-impact geology is already complex, SRM may be used as in this study, to provide evidence on the nature of block rotations and to obtain an improved estimate of the shock point location. In addition, anomalous areas of low SRM may identify allochthonous blocks that have moved inwards more than others. It is possible that the effects of SRM extend for greater distances from the shock point than shock metamorphic features so that deformations producing the most distal features of the crater can be analysed.

Conclusions

A variety of evidence has been presented that a secondary component occurring in the impacted rocks from the central uplift of the Slate Islands crater is a shock remanent magnetization. It is therefore anticipated that a similar halo of magnetically overprinted rocks is present around other impact sites providing the pressure–H_{cr} relations are favourable. Similar studies to the one presented here are thus advocated on craters of different sizes to test the degree of resetting and the rate at which it decays from the shock point. These investigations coupled with data on experimentally-produced SRM may ultimately provide estimates of palaeopressure as well as providing further boundary conditions on models of cratering processes.

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Appendix

DESCRIPTION OF PALAEOMAGNETIC SITES

Site

1, 2 Keweenawan volcanics with irregular 50–55° dip to SW. Geology is complex with strike faulting (see Figs 1, 7A). Large shatter cones up to 1 m in length observed.

4 Three shatter-coned red, amygdular Keweenawan flows.

5 Four shatter-coned Keweenawan flows.

6 Two or three shatter-coned Keweenawan flows exposed on small island, cut by narrow red breccia veins which contain shatter-coned clasts.

7 Diabase dike (width 6 m, dip 70° S strike 80°) intruded into Archean schists and offset dextrally about 30 m along a fault striking 20°. Samples collected from interior and chilled margin of dyke. Well-preserved shatter cones found in fault mylonite and poorer ones in dyke chilled margins.

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8 Severely twisted and sheared diabase dyke. Palaeomagnetic measurements on six samples yield great circles on af cleaning but no consistent orientation. Site rejected.

12, 13 Weakly foliated Archean diorite exhibiting well-preserved shatter cones. NRM is very weak (2×10^{-6} G).

14 Diabase dyke (width 20 m, dip 90°, strike 80°) adjacent to shatter-coned Archean diorite of site 13. Chilled margin samples show small (0.5 cm in length) shatter cones which display rare unweathered surface when rock is broken.

15 Poor, rounded outcrop of irregular diabase intrusion showing a vertical contact with heavily shatter-coned Archean porphyry. Breccia is injected along contact and the diabase also is brecciated into larger fragments along this zone (see Fig. 7B).

19 Olivine diabase dyke (width 10 m, dip 50° S, strike 224°) intruded parallel to foliation of Archean host rock porphyry. Samples collected from interior, chilled margin and baked country rock zone of dike. Shatter cones uncommon but locally well-preserved in chilled margin.

20 Alternate green and pink banded Archean (?) tuff-felsite. Thin breccia dykes cross cut and carry blocks of local country rock. Samples collected from pink blocks and also from red and green host rocks. Green samples, magnetically weak and unstable, were rejected. Well-preserved shatter cones found in the pink tuff/felsite.

23 Diabase dyke (width 4 m, dip 90°, strike 20°). Samples taken from interior and chilled margin. Possible mylonitic material along contact. Poorly-preserved shatter cones occur in chilled margin. Palaeomagnetic data are scattered but could be consistent with a 90° clockwise rotation of dyke about a near vertical axis, needed to remove highly anomalous shatter cone orientation.

24 Shatter-coned carbonated lamprophyre (T. Mitchell 1978, personal communication) with thin green breccia dykes along contact with host rocks. Intrusion magnetically soft and unstable. Breccia stable ($D = 90°$, $I = +60°$) but directions scattered. Local field at time of breccia injection a possible influence as the lamprophyre is strongly magnetic (NRM = 1.4×10^{-2} G). Lamprophyre K–Ar age is 1183 ± 16 Ga (R. Bottomley 1977, personal communication).

25 Diabase dyke (width 2 m, dip 60° S, strike 90°) forming natural breakwater at Slate Islands Lighthouse. Samples taken from interior and chilled margin. Nearest shatter-cone locality 0.5 km to the north in east-trending diabase dyke. Throughout intervening region Archean foliation strikes E–W and dips steeply to south, so it is assumed that there has been no relative rotation between the two areas.

26 Shatter-coned, slightly foliated Archean diorite. Samples oriented using magnetic compass due to overcast weather. Rock weakly magnetic (NRM = 2×10^{-6} G).

27 Diabase/gabbro dyke (width 30 m, dip 90°, strike 110°), showing two phases of intrusion. Samples, collected in 1973 using a magnetic compass, were later supplemented by four more oriented using a sun compass. Declination of 1973 samples appeared to be too large by about 10°. Excellent shatter cones preserved in chilled margins, with circular joint patterns suggestive of giant cones in the more massive dyke interior.

29 Rounded outcrops of diabase dyke (width 2.5 m, dip 90°, strike 156°) cutting Archean diorite. The diabase appears to be part of a much larger and more irregular body. Samples collected from shatter-coned chilled margins. The poor estimate of the secondary component direction derived from convergent remagnetization circles and one Hoffman–Day result (Table 5 and Fig. 2C), when rotated to the breccia direction given by the mean of sites 34–38, tends to remove the anomalous shatter cone direction found at the site.

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31 Single Keweenawan flow dipping about 70° SW, immediately overlying the basal one which appears to be in faulted contact with Archean rock below. There may be up to 10° orientation error as samples were collected in 1973 using a magnetic compass.

32 Shatter-coned Animikean argillite/iron formation (thickness 30 m, dip 40° W, strike 215° using magnetic compass). Chert-magnetite-hematite layers grade upwards into a grey argillite which becomes red in its uppermost 3 m. The red argillite is overlain with apparent concordance by Keweenawan volcanics although their regional SW dip and SE strike suggest an unconformity. Palaeomagnetic samples were collected from all lithologies. Shatter cones, best developed in the argillite, are several cm in length and show fine detail with clear horsetailing. Their orientation was not measured because the rocks are highly magnetic and either poor weather or deep shadow prevented use of the sun compass.

33 Three Keweenawan lava flows, the middle one showing excellent columnar jointing. Shatter-cones inconspicuous.

34 Dense fine-grained red breccia injected along a zone separated site 32 from Keweenawan diabase (site 42). Directions of remanent magnetization may be slightly in error due to the presence of adjacent Animikean sequence, which was highly magnetic (NRM = 2×10^{-2} G), at the time of the breccia injection.

35 Ramifying red breccia veins and dykes injected mostly into Archean rocks, and lying along or adjacent to the faulted (?) contact of the Animikean/Keweenawan sequence with the underlying Archean. Excellent shatter-coned breccia clasts observed.

36 Buff-coloured, fine-grained breccia with scattered Archean blocks up to 10 cm in dimension which appears to rest on Archean basement. Jointing gives breccia crudely layered appearance with a dip of 5° E, strike 20°. Origin of breccia is problematical. One felspar grain, highly shocked to the plagioclase glass maskelynite has been found (R. A. F. Grieve 1976, personal communication).

37 Green breccia dyke, locally with blocks of country rock up to a metre in length. The breccia colour may reflect the nature of the local host rocks. At site 7 a breccia vein cuts across the diabase dyke. It is red within the dyke but becomes green on either side where it traverses the grey-green Archean schistose rocks.

38 Green breccia carrying shatter-coned clasts, some of a purple siltstone quite unlike other rocks on the islands (clasts collected by R. A. F. Grieve, 1975). Clasts resemble the Upper Keweenawan Freda Formation exposed along the south shore of Lake Superior in Michigan.

39 Diabase dyke with pyroxene phenocrysts (width 3 → 0.2 m, dip 70° S, strike 100°). Samples collected from interior, chilled margin and baked contact zone of dyke. Small segment of a large (10 cm) shatter cone observed.

40 Diabase dyke (dip 80° S, strike 105°) intruded parallel to host rock foliation. Dyke interior, chilled margin and baked host rock samples collected using a magnetic compass. An early 1973 site obtained during overcast weather.

41 Diabase dyke (width 10 m, dip 90°, strike 90°). Interior and chilled margin samples taken using a magnetic compass during overcast weather.

42 Diabase dyke or possibly the massive part of an upturned flow (width 2 m, dip 80° S, strike 150°), separated from Animikean sequence to north (site 32) by thin (less than 1 m thick) red breccia zone of site 34. Interior samples only were collected as no chilled margin was observed.

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